1

_FirstScan
SP0

LD
K1006

OUT
Battery Enable
V7633

LD
K106

OUT
X0 Setup
V7634

LD
K1006

OUT
X1 Setup
V7635

OUT
X2 Setup
V7636

OUT
X3 Setup
V7637

Dry Cycle
C320 C323
RST

Has Lock Out
C322
RST

Has Air Assist
C327
SET

Port Setup 1
S770
SET

For HMI Machine Status
Window
Messages
S760
SET

Check for valid login
from HMI
HMI Login
S1770
SET

LD
K0

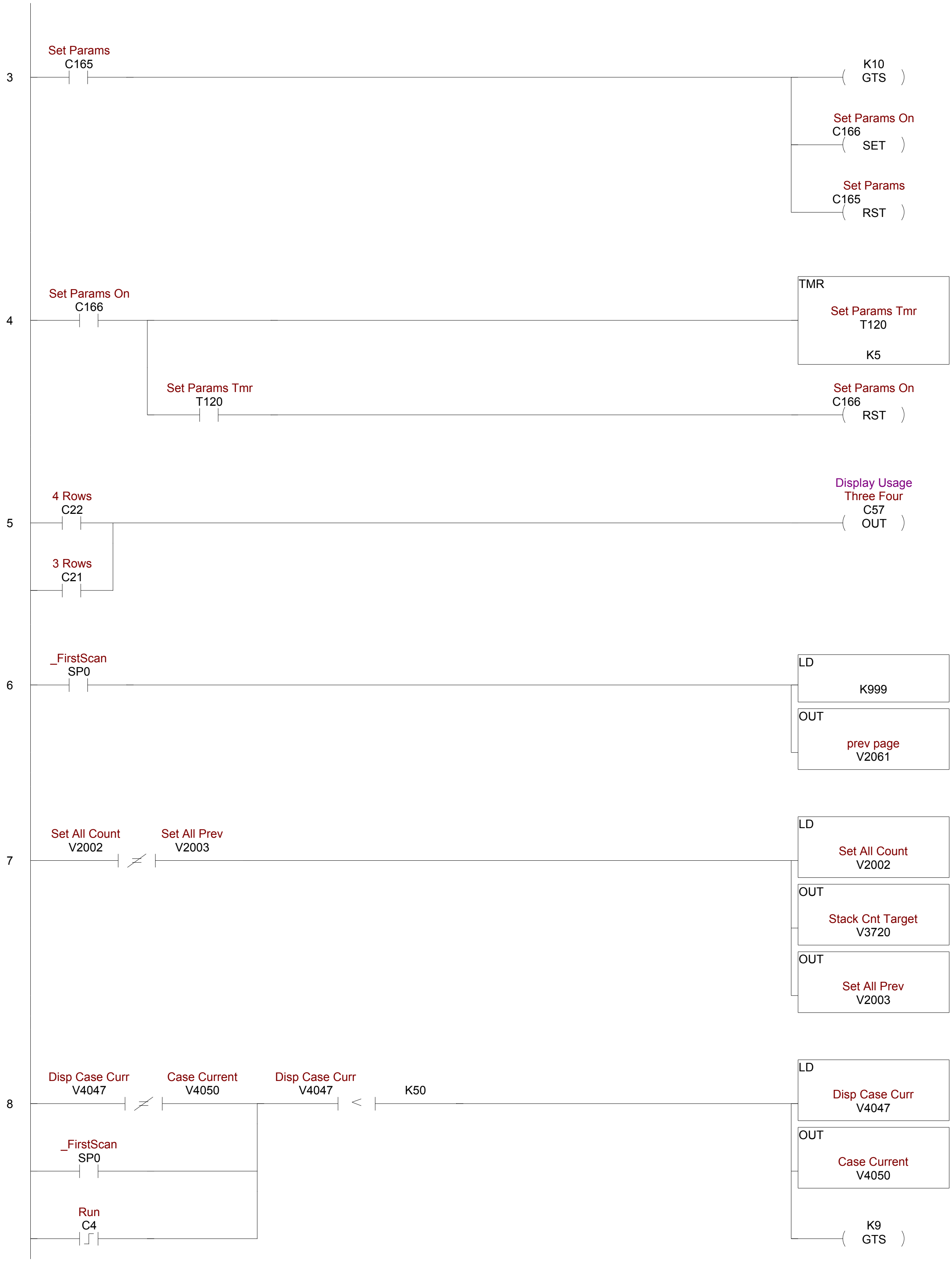
OUT
HMI Login Button
V5010

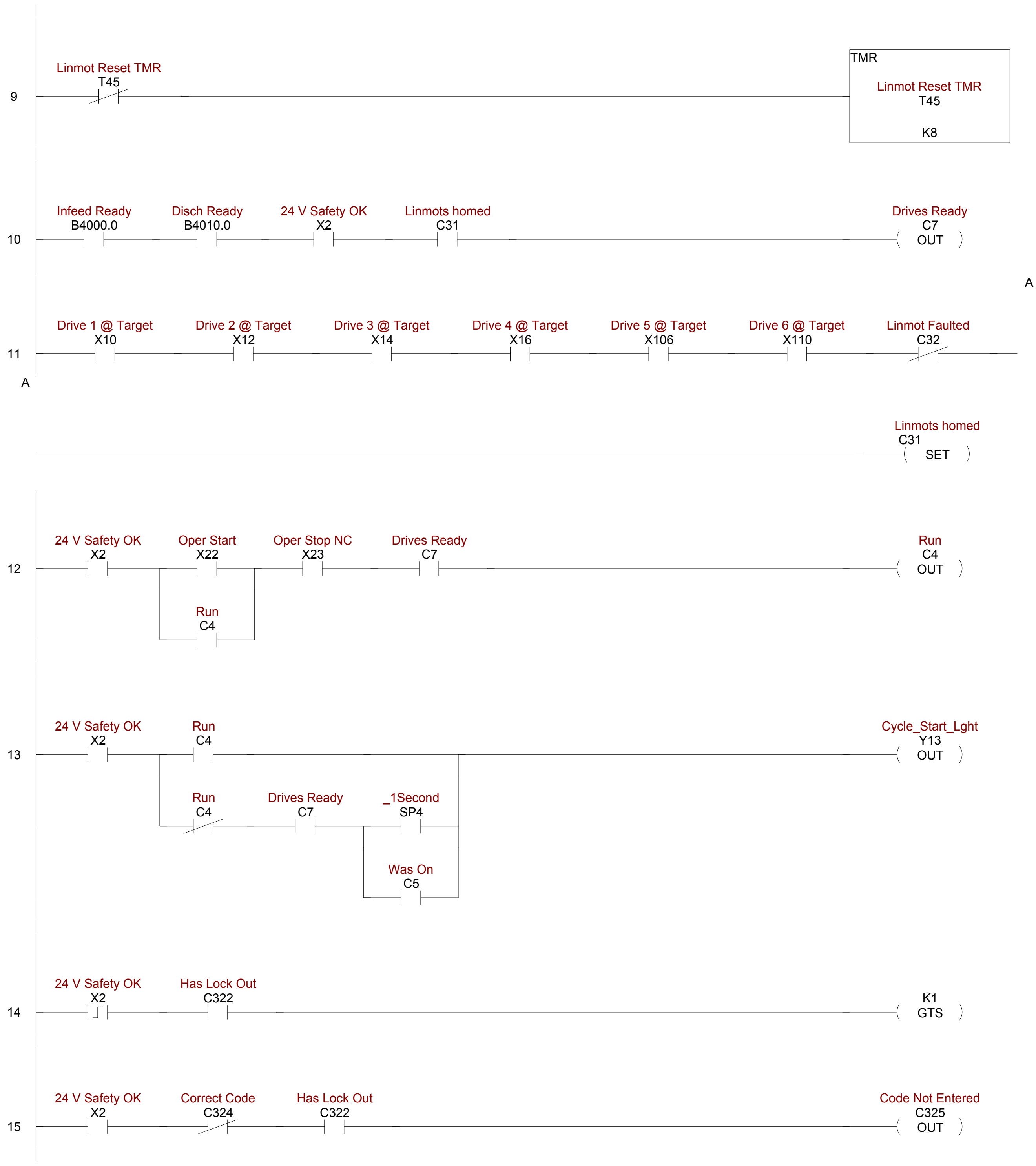
2

Dry Cycle
C320

Run Show Tmr
T111

TMR
Run Show Tmr
T111
K5


3
Set Params
C165
K10
GTS
Set Params On
C166
SET
Set Params
C165
RST
4
Set Params On
C166
TMR
Set Params Tmr
T120
K5
Set Params Tmr
T120
Set Params On
C166
RST
5
4 Rows
C22
3 Rows
C21
Display Usage
Three Four
C57
OUT
6
_FirstScan
SP0
LD
K999
OUT
prev page
V2061
7
Set All Count
V2002
Set All Prev
V2003
LD
Set All Count
V2002
OUT
Stack Cnt Target
V3720
OUT
Set All Prev
V2003
8
Disp Case Curr
V4047
Case Current
V4050
Disp Case Curr
V4047
K50
_FirstScan
SP0
Run
C4
LD
Disp Case Curr
V4047
OUT
Case Current
V4050
K9
GTS


9
Linmot Reset TMR
T45
TMR
Linmot Reset TMR
T45
K8
10
Infeed Ready
B4000.0
Disch Ready
B4010.0
24 V Safety OK
X2
Linmots homed
C31
Drives Ready
C7
OUT
A
11
Drive 1 @ Target
X10
Drive 2 @ Target
X12
Drive 3 @ Target
X14
Drive 4 @ Target
X16
Drive 5 @ Target
X106
Drive 6 @ Target
X110
Linmot Faulted
C32
A
Linmots homed
C31
SET
12
24 V Safety OK
X2
Oper Start
X22
Oper Stop NC
X23
Drives Ready
C7
Run
C4
OUT
Run
C4
13
24 V Safety OK
X2
Run
C4
Cycle_Start_Lght
Y13
OUT
Run
C4
Drives Ready
C7
_1Second
SP4
Was On
C5
14
24 V Safety OK
X2
Has Lock Out
C322
K1
GTS
15
24 V Safety OK
X2
Correct Code
C324
Has Lock Out
C322
Code Not Entered
C325
OUT

16
0 - 100
Step Change Pct
V3723 ≥ K1

LD
Stack Cnt Target
V3720

MUL
0 - 100
Step Change Pct
V3723

DIV
K100

OUT
Step Change Pnt
V4171

Wait for power on - Red Stop Pulled Up - Then time delay for soft start

17
24 V Safety OK
X2

Guard Doors OK
X21

Correct Code
C324

Has Lock Out
C322

Run
C4

TMR
Startup Delay Tm
T31
K30

Startup Delay Tm
T31

Audio_Alarm
Y12
OUT

On start up check the row and diameter selections for errors and set the proper selection bits

ISG
First Step
S0

19
24 V Safety OK
X2

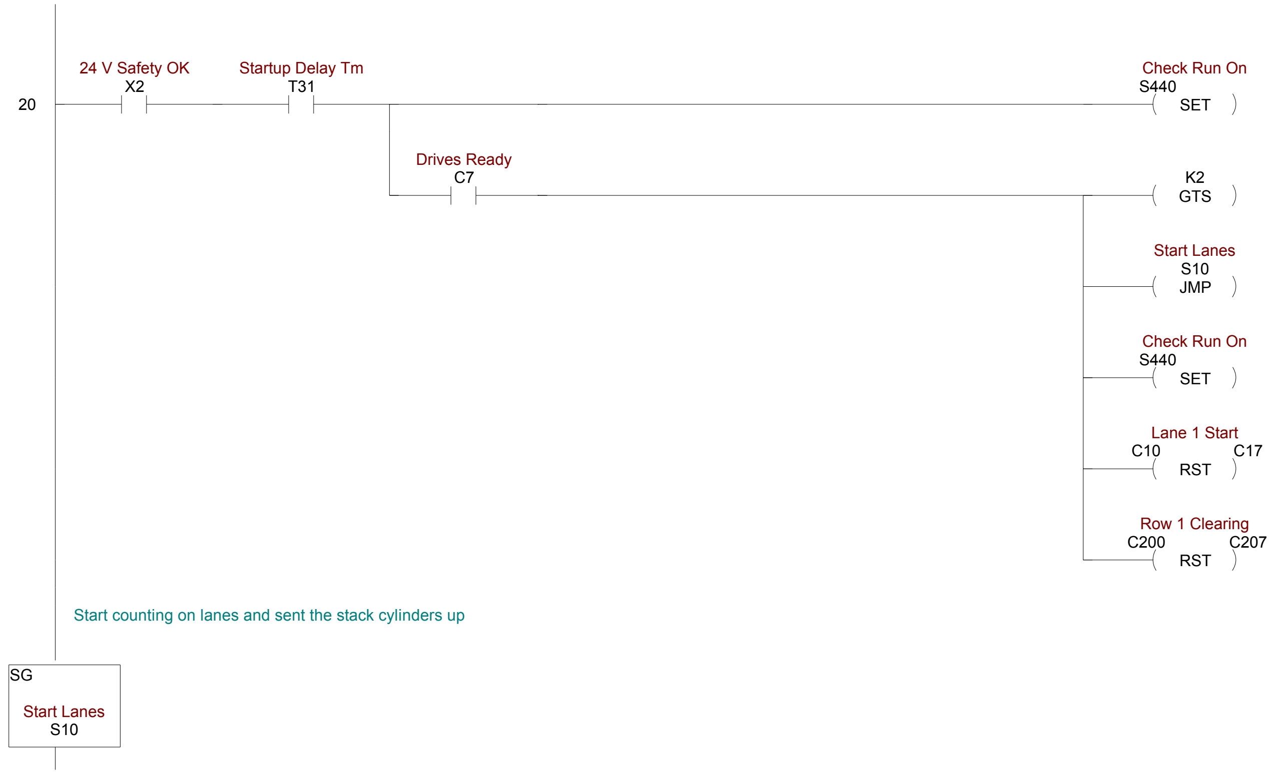
Rate Accum
S300 S377
SET

Twr 1 Lowering
C340 C377
RST

Move Tower 1
C260 C265
RST

Drive 1 Trigger
Y100 Y115
RST

Dry Cycle
C320
RST


20
24 V Safety OK
X2
Startup Delay Tm
T31
Check Run On
S440
SET
Drives Ready
C7
K2
GTS
Start Lanes
S10
JMP
Check Run On
S440
SET
Lane 1 Start
C10
C17
RST
Row 1 Clearing
C200
C207
RST
Start counting on lanes and sent the stack cylinders up
SG
Start Lanes
S10

22

PE Lane 1
S100
(JMP)

Down Lane 1
S102
(JMP)

Lane 1 Go Down
C430
(SET)

PE Lane 2
S110
(JMP)

Down Lane 2
S112
(JMP)

Lane 2 Go Down
C431
(SET)

PE Lane 3
S120
(JMP)

Down Lane 3
S122
(JMP)

Lane 3 Go Down
C432
(SET)

PE Lane 4
S130
(JMP)

Down Lane 4
S132
(JMP)

Lane 4 Go Down
C433
(SET)

Lane 1 Count
CT0 CT7
(RST)

Row 1 PE Blocked
C330 C337
(RST)

Maps input PE's and Down Proxes to bits based on rows selected
Row To Lane
S40
(SET)

Map fork and servo commands to appropriate control based on rows selected
Lane To Row
S170
(SET)

Upper Fork Cntrl
S160
(SET)


Air assist contl
S161
SET
SG
Maps input PE's and
Down Proxes to bits
based on rows selected
Row To Lane
S40
24
PE #2
X4
2 Rows
C20
Lane 1 PE Bit
C400
OUT
PE #1
X3
3 Rows
C21
4 Rows
C22
25
PE #4
X6
2 Rows
C20
Lane 2 PE Bit
C401
OUT
PE #3
X5
3 Rows
C21
PE #2
X4
4 Rows
C22
26
PE #5
X7
3 Rows
C21
Lane 3 PE Bit
C402
OUT
PE #4
X6
4 Rows
C22
27
PE #5
X7
4 Rows
C22
Lane 4 PE Bit
C403
OUT
28
tower 1 down
X100
tower 2 down
X101
tower 3 down
X102
2 Rows
C20
Lane 1 Is Down
C410
OUT
3 Rows
C21
4 Rows
C22

29
tower 4 down X103 — tower 5 down X104 — tower 6 down X105 — 2 Rows C20 — Lane 2 Is Down C411 (OUT)
tower 3 down X102 — tower 4 down X103 — 3 Rows C21
tower 2 down X101 — tower 3 down X102 — 4 Rows C22

30
tower 5 down X104 — tower 6 down X105 — 3 Rows C21 — Lane 3 Is Down C412 (OUT)
tower 4 down X103 — tower 5 down X104 — 4 Rows C22

31
tower 6 down X105 — 4 Rows C22 — Lane 4 Is Down C413 (OUT)

Count individual tortillas for stack count and for rate.

SG
PE Lane 1
S100

33
Lane 1 PE Bit C400 — TMRF SensorMinR1 T150 Sensor Min Tm V2160

34
SensorMinR1 T150 — SensorDebounceR1 T140 (NC) — TMRF SensorDebounceR1 T140 SensorDebounceTm V2161
Debounce On R1 C60 — Debounce On R1 C60 (OUT)

35
Lane 1 PE Bit C400 — Blow Down Enable B3737.0 — Blow Dly 1 S106 (SET)

36
Eye Cnt Ln 1 CTA0 < Step Change Pnt V4171 — LD Step Down Count V3731

37
Eye Cnt Ln 1 CTA0 ≥ Step Change Pnt V4171 — LD Step Down Lower V3724

38
State Not Used _On SP1 — OUT Step Down Lane 1 V4200

When a stack has passed the PE start the travel time and increment the total count.
This stage stays active because more tortillas will be coming by as the last tortilla of a stack moves toward the stacking point.

39
SensorDebounceR1 T140 — Debounce Enable B3737.1 — Row 1 Clearing C200
Lane 1 PE Bit C400 — Debounce Enable B3737.1
Run Show Tmr T111

SGCNT Lane 1 Count CT0 Stack Cnt Target V3720
SGCNT Rate Lane 1 CT10 K9999
SGCNT Row1 By Ctr CT30 Step Down Lane 1 V4200

40
Lane 1 PE Bit C400 — TMRF Ln 1 PE Blk Tmr T20 PE Block Time V2041
Ln 1 PE Blk Tmr T20 — Row 1 PE Blocked C330 (SET)

41
Lane 1 Count CT0 — Row 1 Clearing C200 — Travel Lane 1 S101 (SET)
LDD Lane 1 Accum V2020
ADDD K1
OUTD Lane 1 Accum V2020
Lane 1 Count CT0 (RST)

42
Row 1 Clearing
C200
Travel Lane 1
S101 RST S105
Ind Clr 1
S104 JMP
Lane 1 Go Down
C430 SET
Lane 1 Count
CT0 RST
R1SawPEClear
C210 RST
Resaw Clear 1
C220 RST

Wait for the travel time of the last tortilla of a stack then jump to moving the stack down

SG
Travel Lane 1
S101

44
TMRF
Lane 1 Travel
T0
Travel Ptr
V3733

45
Lane 1 Travel
T0
Down Lane 1
S102 JMP
Lane 1 Go Down
C430 SET
Up lane 1
S103 RST S105

Extend the forks to catch any tortillas which arrive while this stack is being dropped down and moved out. Turn on the down output and release the brake. After the appropriate time (down plus stack move out) transfer back to moving up.

SG
Down Lane 1
S102

the forks are not needed if in a cleanout cycle
wait for a short amount of time after starting the stack down
before turning on the forks

47
TMR
Dwn Frk Dly 1
T40
fork delay
V3734

if in the cleanout process do not jump to the next stage

48
Lane 1 Is Down
C410

row 1 latch
C230

TMRF
Lane 1 Down Time
T1
Tower Down Time
V3735

row 1 latch
C230
OUT

49
Dwn Frk Dly 1
T40

Lane 1 Start
C10

Lower
Lane 1 Forks On
C420
OUT

50
Lane 1 Down Time
T1

Up lane 1
S103
JMP

Lane 1 Go Up
C440
SET

R1SawPEClear
C210
RST

Resaw Clear 1
C220
RST

Move the stack platform back up, with the forks still extended, for the set up time.

SG
Up lane 1
S103

52
TMRF
Lane 1 Up Time
T10
Lanes Up Time
V2040

53
Lane 1 Start
C10

Lower
Lane 1 Forks On
C420
OUT

54
Lane 1 Up Time
T10

Hover Lane 1
S105
JMP

Lane 1 Start
C10
SET

Row 1 Clearing
C200
RST

PE Lane 1
S100
SET

SG
Ind Clr 1
S104

56
Lane 1 Is Down
C410
From HMI
Clear Btn Row 1
C140
R1SawPEClear
C210
SET

57
R1SawPEClear
C210
Lane 1 PE Bit
C400
Resaw Clear 1
C220
SET

58
From HMI
Clear Btn Row 1
C140
Resaw Clear 1
C220
Lane 1 PE Bit
C400
Run
C4
Travel Lane 1
S101
JMP
Row 1 PE Blocked
C330
RST

SG
Hover Lane 1
S105

60
Eye Cnt Ln 1
CTA0
<
Lower
Lower Fork Cnt
V3730
Travel Lane 1
S101
Lower
Lane 1 Forks On
C420
OUT

61
Row1 By Ctr
CT30
Lane 1 Do Step
C450
SET

62
Row1 By Ctr
CT30
Lane 1 Do Step
C450
Lower
Lane 1 Forks On
C420
Row1 By Ctr
CT30
RST

SG
Blow Dly 1
S106

64
TMRF
Blow Delay 1
T160
Blow Dly Time
V3721

65
Blow Delay 1
T160
Blow On 1
S107
JMP

SG
Blow On 1
S107

67
Blow Bit 1
C70
OUT

TMRF
Blow On Time 1
T170
Blow On Time
V3722

68
Blow On Time 1
T170
Blow On 1
S107
RST

SG
PE Lane 2
S110

70
Lane 2 PE Bit
C401

TMRF
SensorMinR2
T151
Sensor Min Tm
V2160

71
SensorMinR2
T151
SensorDebounceR2
T141

Debounce On R2
C61

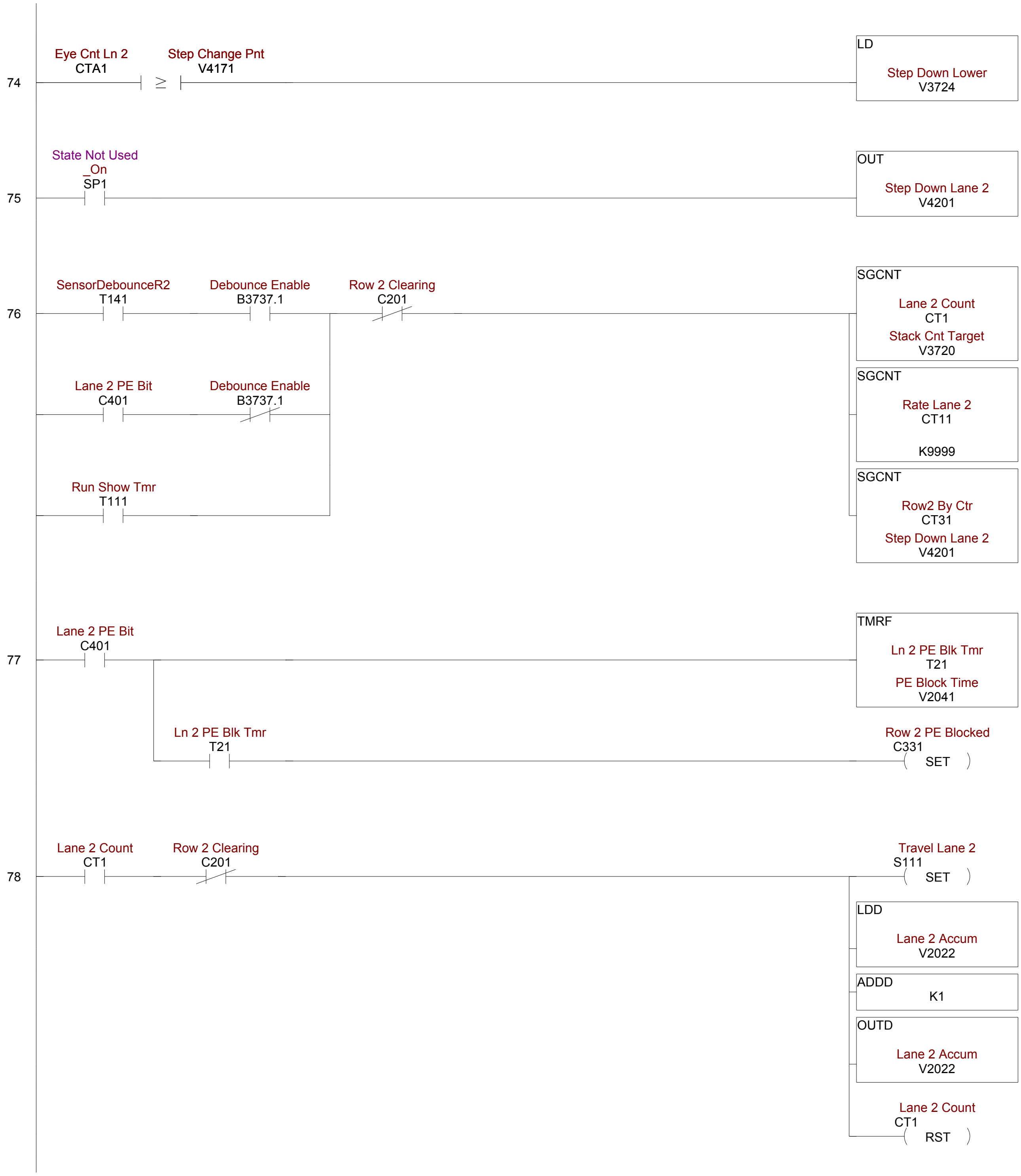
TMRF
SensorDebounceR2
T141
SensorDebounceTm
V2161

Debounce On R2
C61
OUT

72
Lane 2 PE Bit
C401
Blow Down Enable
B3737.0
Blow Dly 2
S116
SET

73
Eye Cnt Ln 2
CTA1
<
Step Change Pnt
V4171

LD
Step Down Count
V3731


74
Eye Cnt Ln 2
CTA1
≥
Step Change Pnt
V4171
LD
Step Down Lower
V3724
75
State Not Used
_On
SP1
OUT
Step Down Lane 2
V4201
76
SensorDebounceR2
T141
Debounce Enable
B3737.1
Row 2 Clearing
C201
Lane 2 PE Bit
C401
Debounce Enable
B3737.1
Run Show Tmr
T111
SGCNT
Lane 2 Count
CT1
Stack Cnt Target
V3720
SGCNT
Rate Lane 2
CT11
K9999
SGCNT
Row2 By Ctr
CT31
Step Down Lane 2
V4201
77
Lane 2 PE Bit
C401
TMRF
Ln 2 PE Blk Tmr
T21
PE Block Time
V2041
Ln 2 PE Blk Tmr
T21
Row 2 PE Blocked
C331
SET
78
Lane 2 Count
CT1
Row 2 Clearing
C201
Travel Lane 2
S111
SET
LDD
Lane 2 Accum
V2022
ADDD
K1
OUTD
Lane 2 Accum
V2022
Lane 2 Count
CT1
RST

79 Row 2 Clearing C201

Travel Lane 2 S111 S115 RST

Ind Clr 2 S114 JMP

Lane 2 Go Down C431 SET

Lane 2 Count CT1 RST

R2SawPEClear C211 RST

Resaw Clear 2 C221 RST

SG Travel Lane 2 S111

81 TMRF Lane 2 Travel T2 Travel Ptr V3733

82 Lane 2 Travel T2

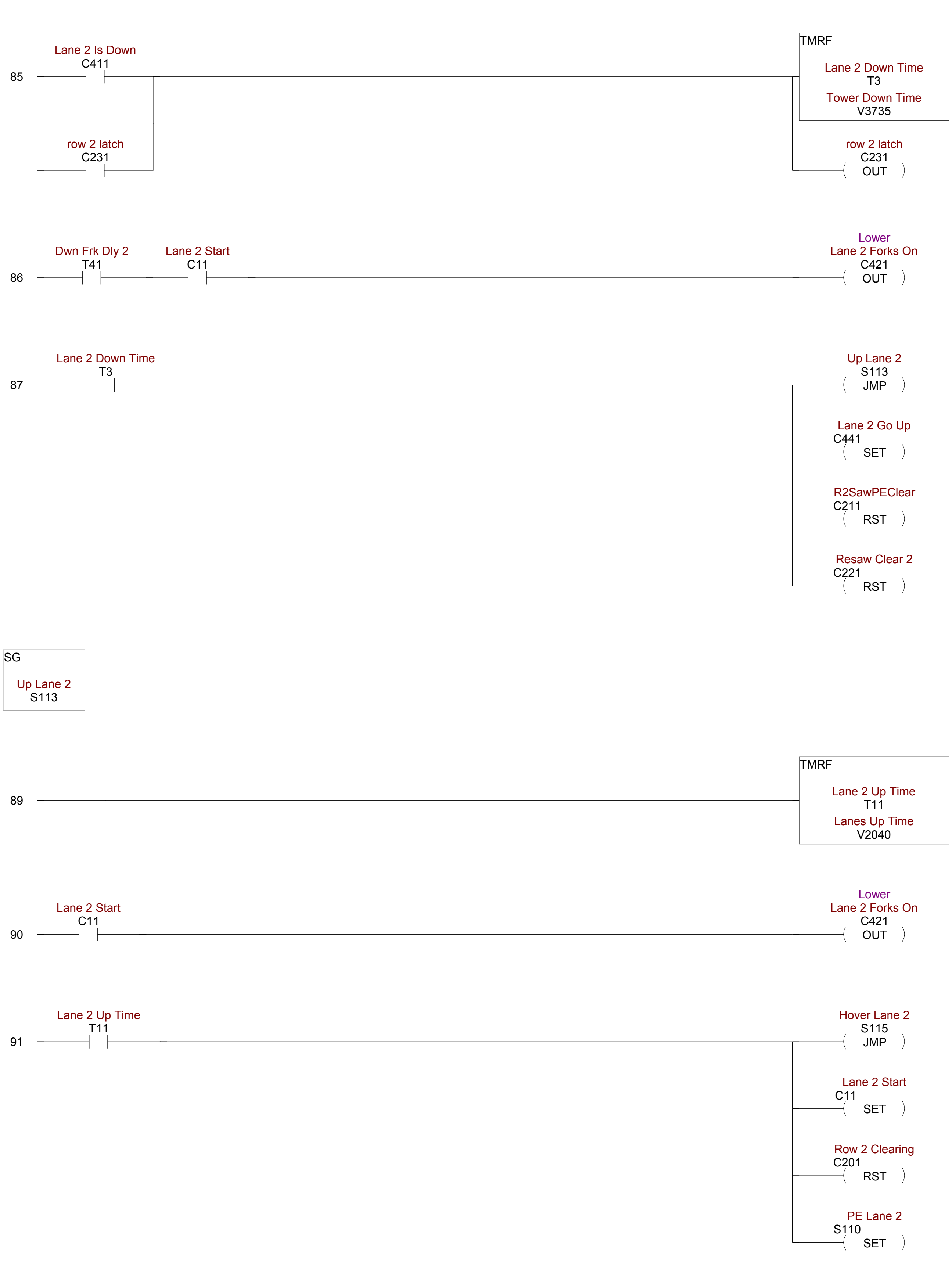
Down Lane 2 S112 JMP

Lane 2 Go Down C431 SET

Up Lane 2 S113 S115 RST

SG Down Lane 2 S112

84 TMR Dwn Frk Dly 2 T41 fork delay V3734


85
Lane 2 Is Down
C411
row 2 latch
C231
TMRF
Lane 2 Down Time
T3
Tower Down Time
V3735
row 2 latch
C231
OUT
86
Dwn Frk Dly 2
T41
Lane 2 Start
C11
Lower
Lane 2 Forks On
C421
OUT
87
Lane 2 Down Time
T3
Up Lane 2
S113
JMP
Lane 2 Go Up
C441
SET
R2SawPEClear
C211
RST
Resaw Clear 2
C221
RST
SG
Up Lane 2
S113
89
TMRF
Lane 2 Up Time
T11
Lanes Up Time
V2040
90
Lane 2 Start
C11
Lower
Lane 2 Forks On
C421
OUT
91
Lane 2 Up Time
T11
Hover Lane 2
S115
JMP
Lane 2 Start
C11
SET
Row 2 Clearing
C201
RST
PE Lane 2
S110
SET

SG
Ind Clr 2
S114

93
Lane 2 Is Down C411 — From HMI Clear Btn Row 2 C141 — R2SawPEClear C211 (SET)

94
R2SawPEClear C211 — Lane 2 PE Bit C401 — Resaw Clear 2 C221 (SET)

95
From HMI Clear Btn Row 2 C141 — Resaw Clear 2 C221 — Lane 2 PE Bit C401 — Run C4 — Travel Lane 2 S111 (JMP)
Row 2 PE Blocked C331 (RST)

SG
Hover Lane 2
S115

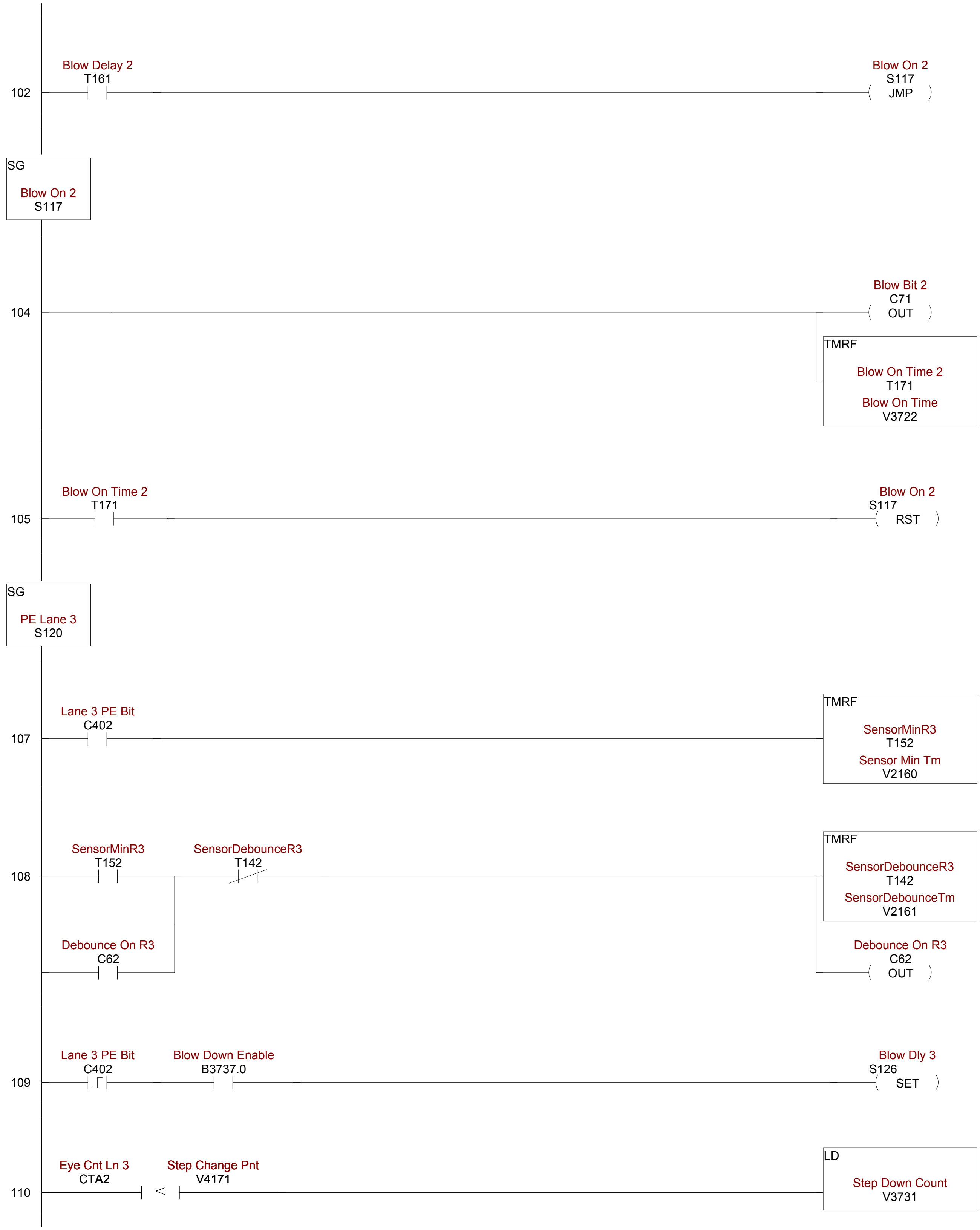
97
Eye Cnt Ln 2 CTA1 < Lower Lower Fork Cnt V3730 — Travel Lane 2 S111 — Lower Lane 2 Forks On C421 (OUT)

98
Row2 By Ctr CT31 — Lane 2 Do Step C451 (SET)

99
Row2 By Ctr CT31 — Lane 2 Do Step C451 — Row2 By Ctr CT31 (RST)
Lower Lane 2 Forks On C421

SG
Blow Dly 2
S116

101
TMRF
Blow Delay 2
T161
Blow Dly Time
V3721


102
Blow Delay 2
T161
Blow On 2
S117
JMP
SG
Blow On 2
S117
104
Blow Bit 2
C71
OUT
TMRF
Blow On Time 2
T171
Blow On Time
V3722
105
Blow On Time 2
T171
Blow On 2
S117
RST
SG
PE Lane 3
S120
107
Lane 3 PE Bit
C402
TMRF
SensorMinR3
T152
Sensor Min Tm
V2160
108
SensorMinR3
T152
SensorDebounceR3
T142
TMRF
SensorDebounceR3
T142
SensorDebounceTm
V2161
Debounce On R3
C62
Debounce On R3
C62
OUT
109
Lane 3 PE Bit
C402
Blow Down Enable
B3737.0
Blow Dly 3
S126
SET
110
Eye Cnt Ln 3
CTA2
<
Step Change Pnt
V4171
LD
Step Down Count
V3731

111
Eye Cnt Ln 3 CTA2 ≥ Step Change Pnt V4171
LD Step Down Lower V3724

112
State Not Used _On SP1
OUT Step Down Lane 3 V4202

113
SensorDebounceR3 T142 — Debounce Enable B3737.1 — Row 3 Clearing C202
Lane 3 PE Bit C402 — Debounce Enable B3737.1
Run Show Tmr T111
SGCNT Lane 3 Count CT2 Stack Cnt Target V3720
SGCNT Rate Lane 3 CT12 K9999
SGCNT Row3 By Ctr CT32 Step Down Lane 3 V4202

114
Lane 3 PE Bit C402
TMRF Ln 3 PE Blk Tmr T22 PE Block Time V2041
Ln 3 PE Blk Tmr T22
Row 3 PE Blocked C332 (SET)

115
Lane 3 Count CT2 — Row 3 Clearing C202
Travel Lane 3 S121 (SET)
LDD Lane 3 Accum V2024
ADDD K1
OUTD Lane 3 Accum V2024
Lane 3 Count CT2 (RST)

116 — Row 3 Clearing C202 (contact)
- Travel Lane 3 S121 S125 (RST)
- Ind Clr 3 S124 (JMP)
- Lane 3 Go Down C432 (SET)
- Lane 3 Count CT2 (RST)
- R3SawPEClear C212 (RST)
- Resaw Clear 3 C222 (RST)

SG
Travel Lane 3
S121

118 — TMRF
Lane 3 Travel
T4
Travel Ptr
V3733

119 — Lane 3 Travel T4 (contact)
- Down Lane 3 S122 (JMP)
- Lane 3 Go Down C432 (SET)
- Up Lane 3 S123 S125 (RST)

SG
Down Lane 3
S122

121 — TMR
Dwn Frk Dly 3
T42
fork delay
V3734

122
Lane 3 Is Down C412
row 3 latch C232
TMRF Lane 3 Down Time T5 Tower Down Time V3735
row 3 latch C232 OUT

123
Dwn Frk Dly 3 T42
Lane 3 Start C12
Lower Lane 3 Forks On C422 OUT

124
Lane 3 Down Time T5
Up Lane 3 S123 JMP
Lane 3 Go Up C442 SET
R3SawPEClear C212 RST
Resaw Clear 3 C222 RST

SG Up Lane 3 S123

126
TMRF Lane 3 Up Time T12 Lanes Up Time V2040

127
Lane 3 Start C12
Lower Lane 3 Forks On C422 OUT

128
Lane 3 Up Time T12
Hover Lane 3 S125 JMP
Lane 3 Start C12 SET
Row 3 Clearing C202 RST
PE Lane 3 S120 SET

SG
Ind Clr 3
S124

130
Lane 3 Is Down
C412
From HMI
Clear Btn Row 3
C142
R3SawPEClear
C212
SET

131
R3SawPEClear
C212
Lane 3 PE Bit
C402
Resaw Clear 3
C222
SET

132
From HMI
Clear Btn Row 3
C142
Resaw Clear 3
C222
Lane 3 PE Bit
C402
Run
C4
Travel Lane 3
S121
JMP
Row 3 PE Blocked
C332
RST

SG
Hover Lane 3
S125

134
Eye Cnt Ln 3
CTA2
<
Lower
Lower Fork Cnt
V3730
Travel Lane 3
S121
Lower
Lane 3 Forks On
C422
OUT

135
Row3 By Ctr
CT32
Lane 3 Do Step
C452
SET

136
Row3 By Ctr
CT32
Lane 3 Do Step
C452
Lower
Lane 3 Forks On
C422
Row3 By Ctr
CT32
RST

SG
Blow Dly 3
S126

138
TMRF
Blow Delay 3
T162
Blow Dly Time
V3721

139
Blow Delay 3
T162
Blow On 3
S127
JMP

SG
Blow On 3
S127

141
Blow Bit 3
C72
OUT

TMRF
Blow On Time 3
T172
Blow On Time
V3722

142
Blow On Time 3
T172
Blow On 3
S127
RST

SG
PE Lane 4
S130

144
Lane 4 PE Bit
C403
TMRF
SensorMinR4
T153
Sensor Min Tm
V2160

145
SensorMinR4
T153
SensorDebounceR4
T143
TMRF
SensorDebounceR4
T143
SensorDebounceTm
V2161

Debounce On R4
C63
Debounce On R4
C63
OUT

146
Lane 4 PE Bit
C403
Blow Down Enable
B3737.0
Blow Dly 4
S136
SET

147
Eye Cnt Ln 4
CTA3
<
Step Change Pnt
V4171
LD
Step Down Count
V3731

148
Eye Cnt Ln 4 CTA3 ≥ Step Change Pnt V4171
LD Step Down Lower V3724

149
State Not Used _On SP1
OUT Step Down Lane 4 V4203

150
SensorDebounceR4 T143 — Debounce Enable B3737.1 — Row 4 Clearing C203
Lane 4 PE Bit C403 — Debounce Enable B3737.1
Run Show Tmr T111
SGCNT Lane 4 Count CT3 Stack Cnt Target V3720
SGCNT Rate Lane 4 CT13 K9999
SGCNT Row4 By Ctr CT33 Step Down Lane 4 V4203

151
Lane 4 PE Bit C403
TMRF Ln 4 PE Blk Tmr T23 PE Block Time V2041
Ln 4 PE Blk Tmr T23
Row 4 PE Blocked C333 (SET)

152
Lane 4 Count CT3 — Row 4 Clearing C203
Travel Lane 4 S131 (SET)
LDD Lane 4 Accum V2026
ADDD K1
OUTD Lane 4 Accum V2026
Lane 4 Count CT3 (RST)

153 Row 4 Clearing C203 —| |—
- Travel Lane 4 S131 S135 (RST)
- Ind Clr 4 S134 (JMP)
- Lane 4 Go Down C433 (SET)
- Lane 4 Count CT3 (RST)
- R4SawPEClear C213 (RST)
- Resaw Clear 4 C223 (RST)

SG
Travel Lane 4
S131

155
TMRF
Lane 4 Travel
T6
Travel Ptr
V3733

156 Lane 4 Travel T6 —| |—
- Down Lane 4 S132 (JMP)
- Lane 4 Go Down C433 (SET)
- Up Lane 4 S133 S135 (RST)

SG
Down Lane 4
S132

158
TMR
Dwn Frk Dly 4
T43
fork delay
V3734

159
Lane 4 Is Down C413 — TMRF Lane 4 Down Time T7 Tower Down Time V3735
row 4 latch C233 — row 4 latch C233 OUT

160
Dwn Frk Dly 4 T43 — Lane 4 Start C13 — Lower Lane 4 Forks On C423 OUT

161
Lane 4 Down Time T7 — Up Lane 4 S133 JMP
Lane 4 Go Up C443 SET
R4SawPEClear C213 RST
Resaw Clear 4 C223 RST

SG
Up Lane 4
S133

163
TMRF Lane 4 Up Time T13 Lanes Up Time V2040

164
Lane 4 Start C13 — Lower Lane 4 Forks On C423 OUT

165
Lane 4 Up Time T13 — Hover Lane 4 S135 JMP
Lane 4 Start C13 SET
Row 4 Clearing C203 RST
PE Lane 4 S130 SET

SG
Ind Clr 4
S134

167
Lane 4 Is Down C413 — From HMI Clear Btn Row 4 C143 — R4SawPEClear C213 (SET)

168
R4SawPEClear C213 — Lane 4 PE Bit C403 — Resaw Clear 4 C223 (SET)

169
From HMI Clear Btn Row 4 C143 — Resaw Clear 4 C223 — Lane 4 PE Bit C403 — Run C4 — Travel Lane 4 S131 (JMP)
Row 4 PE Blocked C333 (RST)

SG
Hover Lane 4
S135

171
Eye Cnt Ln 4 CTA3 < Lower Lower Fork Cnt V3730 — Travel Lane 4 S131 — Lower Lane 4 Forks On C423 (OUT)

172
Row4 By Ctr CT33 — Lane 4 Do Step C453 (SET)

173
Row4 By Ctr CT33 — Lane 4 Do Step C453 — Row4 By Ctr CT33 (RST)
Lower Lane 4 Forks On C423

SG
Blow Dly 4
S136

175
TMRF
Blow Delay 4
T163
Blow Dly Time
V3721

176 Blow Delay 4 T163 —| |— Blow On 4 S137 (JMP)

SG
Blow On 4
S137

178 Blow Bit 4 C73 (OUT)
TMRF
Blow On Time 4
T173
Blow On Time
V3722

179 Blow On Time 4 T173 —| |— Blow On 4 S137 (RST)

SG
Upper Fork Cntrl
S160

181 Down Lane 1 S102 —| |—
TMRF
Upper Frk Dly 1
T100
Fork Delay Upr
V3741

the forks are not needed if in a cleanout cycle
wait for a short amount of time after starting the stack down
before turning on the forks

182 Eye Cnt Ln 1 CTA0 < Upper Fork Cnt V3740 — Travel Lane 1 S101 —|/|— Down Lane 1 S102 —|/|— Ind Clr 1 S104 —|/|— Upper Lane_1 Forks On C460 (OUT)
Down Lane 1 S102 —| |— Upper Frk Dly 1 T100 —| |—

183 Down Lane 2 S112 —| |—
TMRF
Upper Frk Dly 2
T101
Fork Delay Upr
V3741

the forks are not needed if in a cleanout cycle
wait for a short amount of time after starting the stack down
before turning on the forks

184
Eye Cnt Ln 2 CTA1 < Upper Fork Cnt V3740 — Travel Lane 2 S111 (NC) — Down Lane 2 S112 (NC) — Ind Clr 2 S114 (NC) — Upper Lane_2 Forks On C461 OUT
Down Lane 2 S112 — Upper Frk Dly 2 T101

185
Down Lane 3 S122 — TMRF Upper Frk Dly 3 T102 Fork Delay Upr V3741

the forks are not needed if in a cleanout cycle
wait for a short amount of time after starting the stack down
before turning on the forks

186
Eye Cnt Ln 3 CTA2 < Upper Fork Cnt V3740 — Travel Lane 3 S121 (NC) — Down Lane 3 S122 (NC) — Ind Clr 3 S124 (NC) — Upper Lane_3 Forks On C462 OUT
Down Lane 3 S122 — Upper Frk Dly 3 T102

187
Down Lane 4 S132 — TMRF Upper Frk Dly 4 T103 Fork Delay Upr V3741

the forks are not needed if in a cleanout cycle
wait for a short amount of time after starting the stack down
before turning on the forks

188
Eye Cnt Ln 4 CTA3 < Upper Fork Cnt V3740 — Travel Lane 4 S131 (NC) — Down Lane 4 S132 (NC) — Ind Clr 4 S134 (NC) — Upper Lane_4 Forks On C463 OUT
Down Lane 4 S132 — Upper Frk Dly 4 T103

SG
Air assist contl
S161

190
Blow Bit 1 C70 — 2 Rows C20 — Two Row air assist Lane 1 Air Assist 2 Y15 OUT

191
Blow Bit 2 C71 — 2 Rows C20 — Two Row air assist Lane 2 Air Assist 5 Y120 OUT

192
Blow Bit 1 C70 — 4 Rows C22 — Four Row air assist Lane 1 Air Assist 1 Y14 OUT

193
Blow Bit 2 C71 — 4 Rows C22 — Four Row air assist Lane 2 Air Assist 3 Y16 OUT

194
Blow Bit 3 C72 — 4 Rows C22 — Four Row air assist Lane 3 Air Assist 4 Y17 OUT

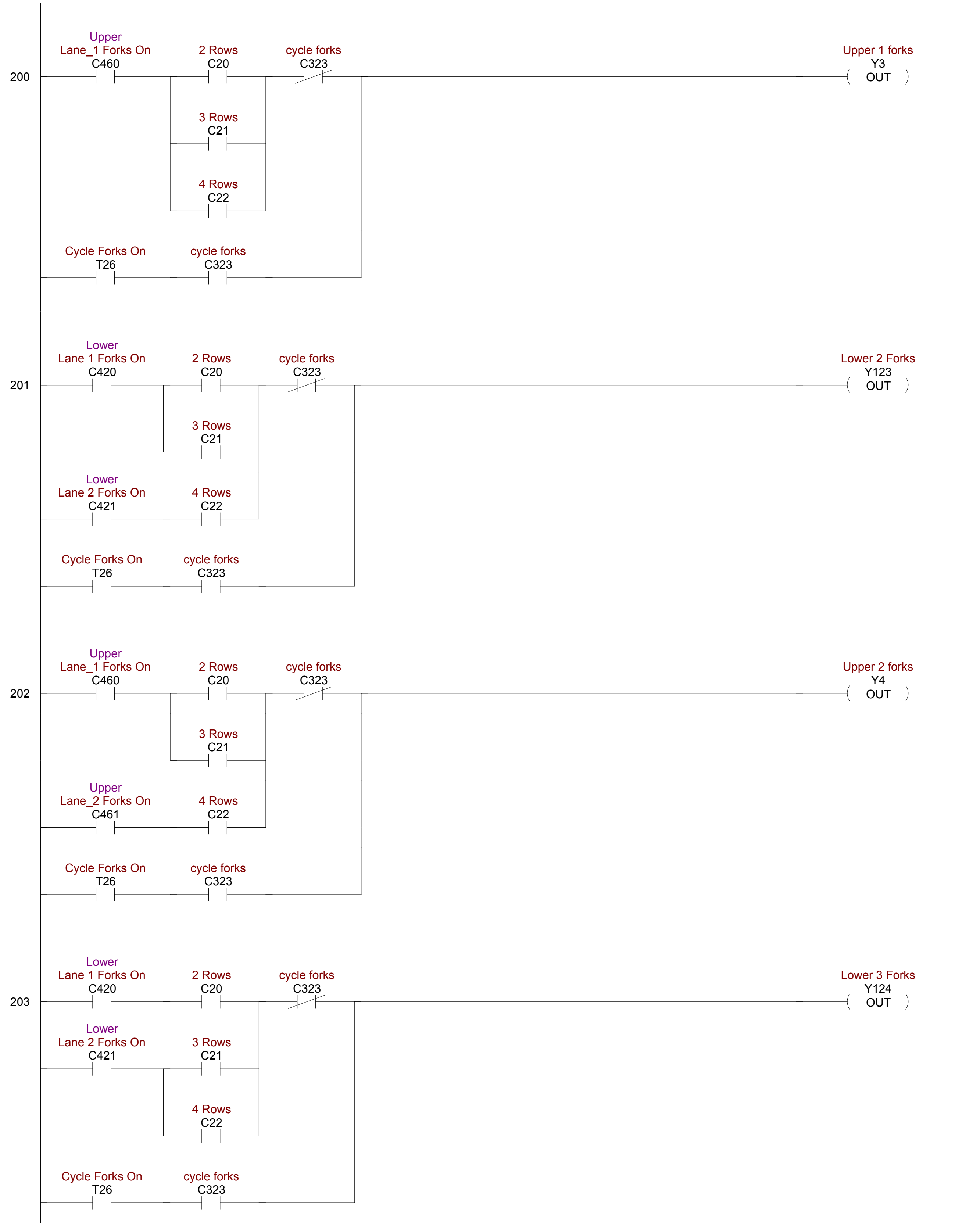
195
Blow Bit 4 C73 — 4 Rows C22 — Four Row air assist Lane 4 Air Assist 6 Y121 OUT

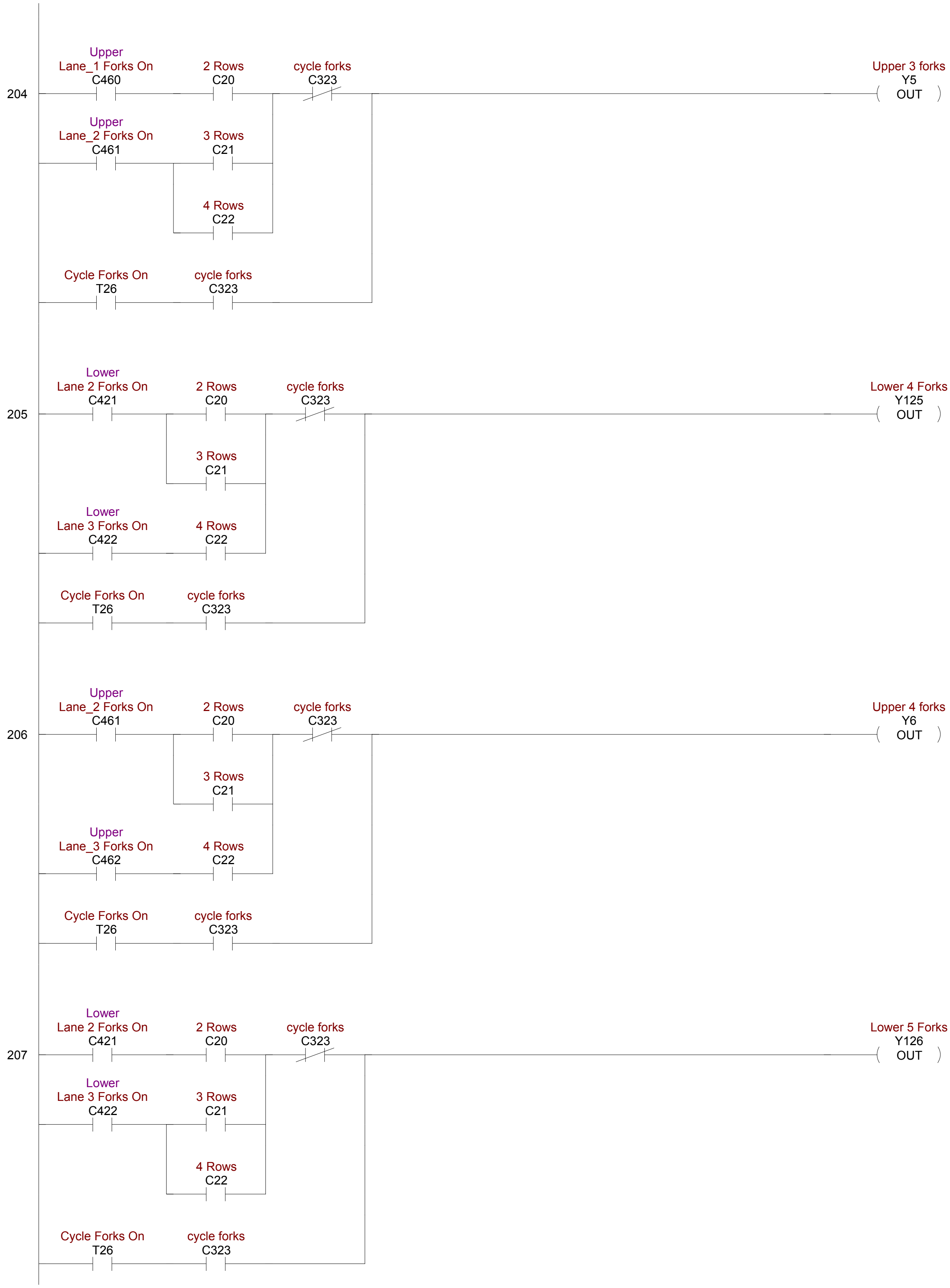
SG
Map fork and servo commands to appropriate control based on rows selected
Lane To Row
S170

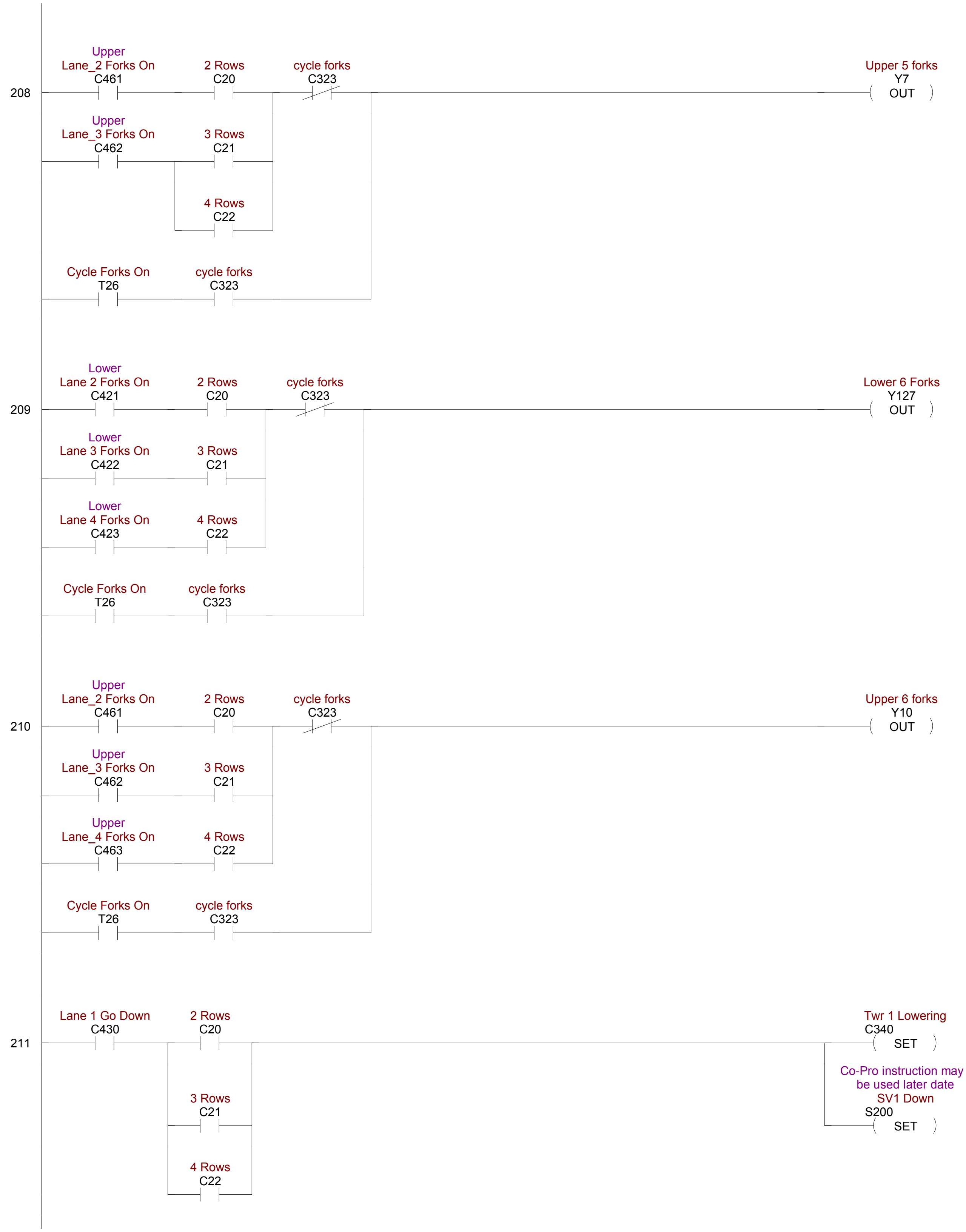
197
cycle forks C323 — Cycle Forks Off T27 — TMR Cycle Forks On T26 K25

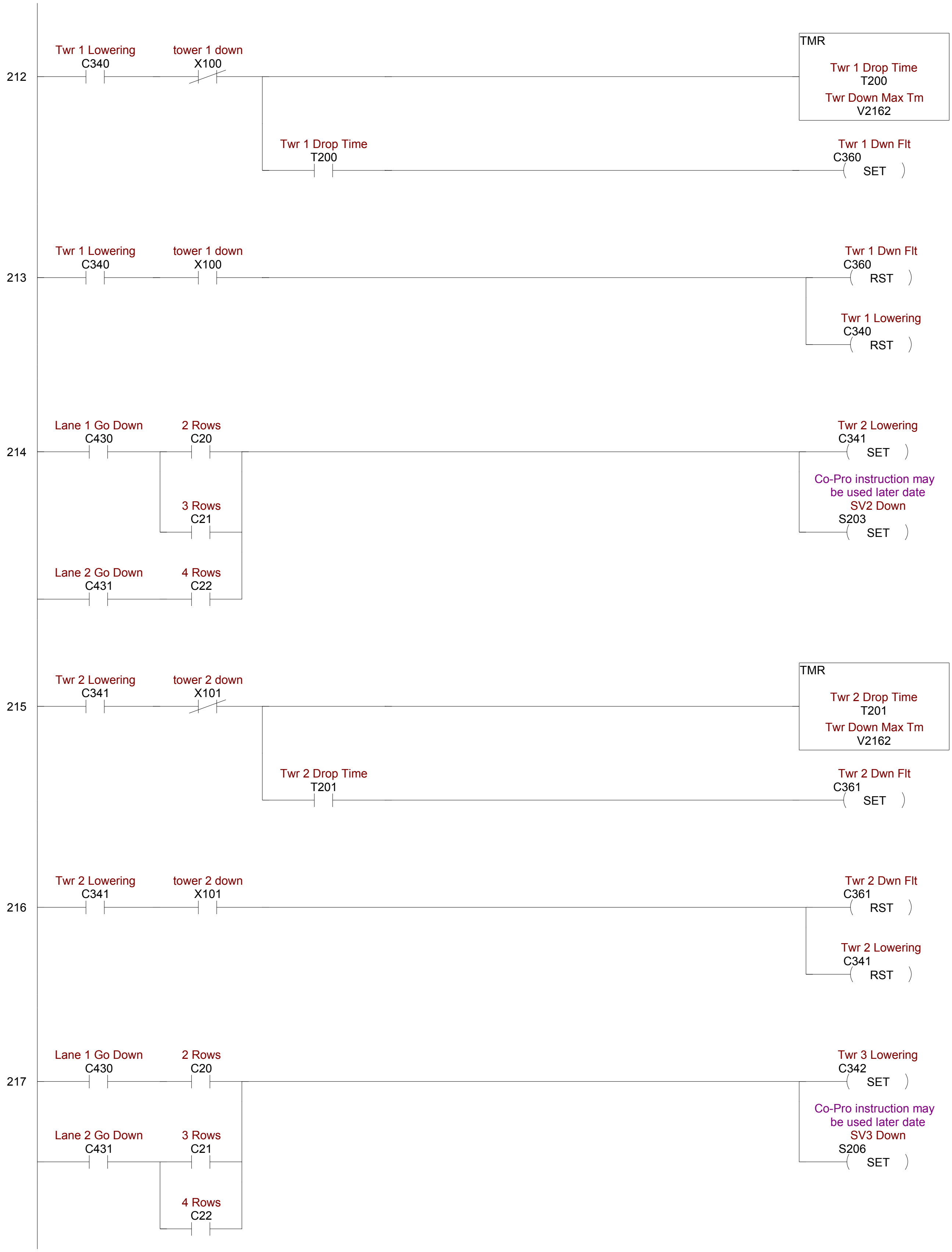
198
Cycle Forks On T26 — TMR Cycle Forks Off T27 K25

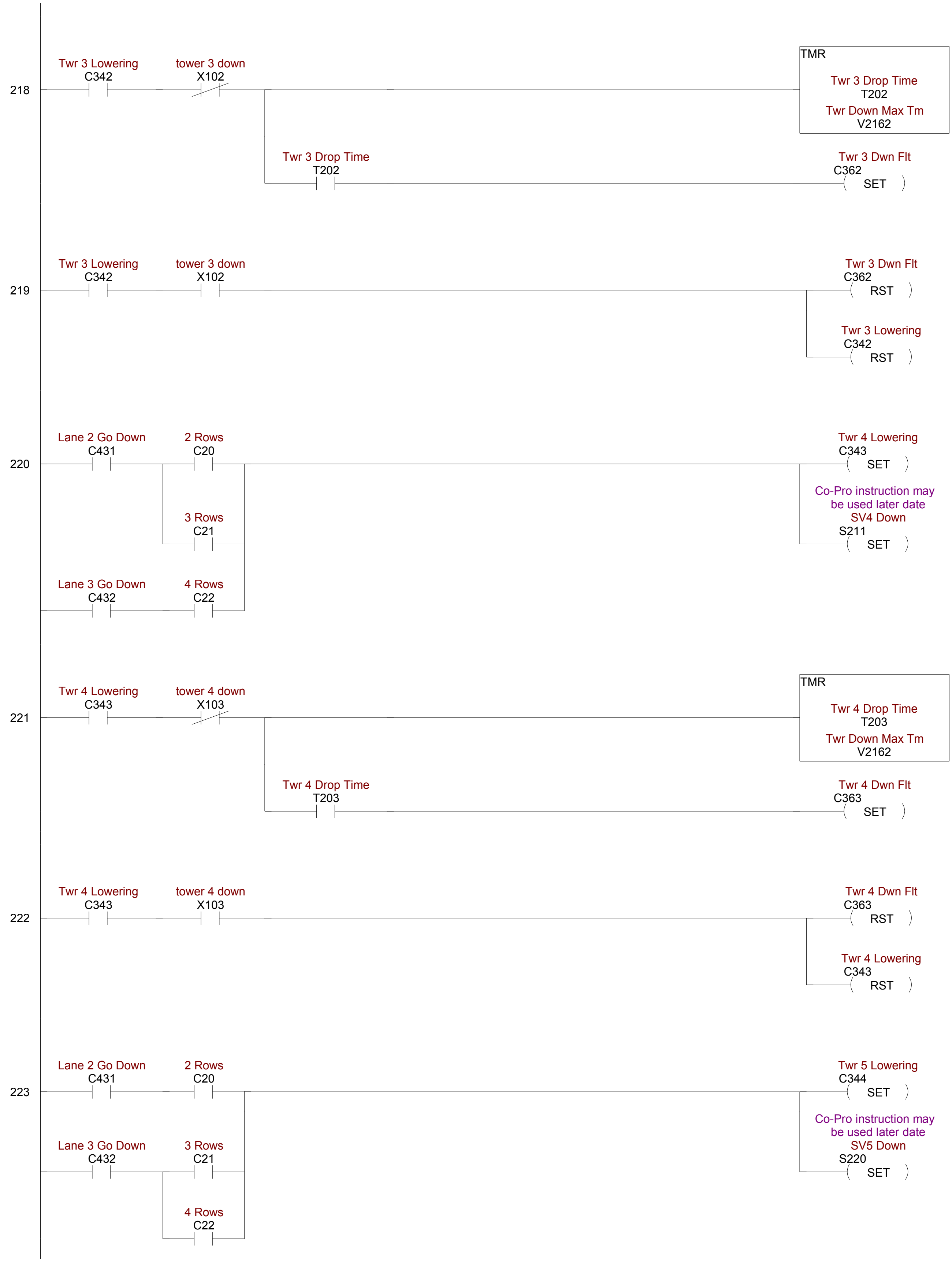
199
Lower Lane 1 Forks On C420 — 2 Rows C20 / 3 Rows C21 / 4 Rows C22 — cycle forks C323 — Lower 1 Forks Y122 OUT
Cycle Forks On T26 — cycle forks C323


200
Upper
Lane_1 Forks On
C460
2 Rows
C20
cycle forks
C323
Upper 1 forks
Y3
OUT
3 Rows
C21
4 Rows
C22
Cycle Forks On
T26
cycle forks
C323
201
Lower
Lane 1 Forks On
C420
2 Rows
C20
cycle forks
C323
Lower 2 Forks
Y123
OUT
3 Rows
C21
Lower
Lane 2 Forks On
C421
4 Rows
C22
Cycle Forks On
T26
cycle forks
C323
202
Upper
Lane_1 Forks On
C460
2 Rows
C20
cycle forks
C323
Upper 2 forks
Y4
OUT
3 Rows
C21
Upper
Lane_2 Forks On
C461
4 Rows
C22
Cycle Forks On
T26
cycle forks
C323
203
Lower
Lane 1 Forks On
C420
2 Rows
C20
cycle forks
C323
Lower 3 Forks
Y124
OUT
Lower
Lane 2 Forks On
C421
3 Rows
C21
4 Rows
C22
Cycle Forks On
T26
cycle forks
C323


204
Upper Lane_1 Forks On C460
2 Rows C20
cycle forks C323
Upper 3 forks Y5 OUT
Upper Lane_2 Forks On C461
3 Rows C21
4 Rows C22
Cycle Forks On T26
cycle forks C323
205
Lower Lane 2 Forks On C421
2 Rows C20
cycle forks C323
Lower 4 Forks Y125 OUT
3 Rows C21
Lower Lane 3 Forks On C422
4 Rows C22
Cycle Forks On T26
cycle forks C323
206
Upper Lane_2 Forks On C461
2 Rows C20
cycle forks C323
Upper 4 forks Y6 OUT
3 Rows C21
Upper Lane_3 Forks On C462
4 Rows C22
Cycle Forks On T26
cycle forks C323
207
Lower Lane 2 Forks On C421
2 Rows C20
cycle forks C323
Lower 5 Forks Y126 OUT
Lower Lane 3 Forks On C422
3 Rows C21
4 Rows C22
Cycle Forks On T26
cycle forks C323


208
Upper
Lane_2 Forks On
C461
2 Rows
C20
cycle forks
C323
Upper 5 forks
Y7
OUT
Upper
Lane_3 Forks On
C462
3 Rows
C21
4 Rows
C22
Cycle Forks On
T26
cycle forks
C323
209
Lower
Lane 2 Forks On
C421
2 Rows
C20
cycle forks
C323
Lower 6 Forks
Y127
OUT
Lower
Lane 3 Forks On
C422
3 Rows
C21
Lower
Lane 4 Forks On
C423
4 Rows
C22
Cycle Forks On
T26
cycle forks
C323
210
Upper
Lane_2 Forks On
C461
2 Rows
C20
cycle forks
C323
Upper 6 forks
Y10
OUT
Upper
Lane_3 Forks On
C462
3 Rows
C21
Upper
Lane_4 Forks On
C463
4 Rows
C22
Cycle Forks On
T26
cycle forks
C323
211
Lane 1 Go Down
C430
2 Rows
C20
Twr 1 Lowering
C340
SET
Co-Pro instruction may
be used later date
SV1 Down
S200
SET
3 Rows
C21
4 Rows
C22


212
Twr 1 Lowering
C340
tower 1 down
X100
TMR
Twr 1 Drop Time
T200
Twr Down Max Tm
V2162
Twr 1 Drop Time
T200
Twr 1 Dwn Flt
C360
SET
213
Twr 1 Lowering
C340
tower 1 down
X100
Twr 1 Dwn Flt
C360
RST
Twr 1 Lowering
C340
RST
214
Lane 1 Go Down
C430
2 Rows
C20
3 Rows
C21
Lane 2 Go Down
C431
4 Rows
C22
Twr 2 Lowering
C341
SET
Co-Pro instruction may
be used later date
SV2 Down
S203
SET
215
Twr 2 Lowering
C341
tower 2 down
X101
TMR
Twr 2 Drop Time
T201
Twr Down Max Tm
V2162
Twr 2 Drop Time
T201
Twr 2 Dwn Flt
C361
SET
216
Twr 2 Lowering
C341
tower 2 down
X101
Twr 2 Dwn Flt
C361
RST
Twr 2 Lowering
C341
RST
217
Lane 1 Go Down
C430
2 Rows
C20
Lane 2 Go Down
C431
3 Rows
C21
4 Rows
C22
Twr 3 Lowering
C342
SET
Co-Pro instruction may
be used later date
SV3 Down
S206
SET


218
Twr 3 Lowering C342
tower 3 down X102
TMR
Twr 3 Drop Time T202
Twr Down Max Tm V2162
Twr 3 Drop Time T202
Twr 3 Dwn Flt C362 SET
219
Twr 3 Lowering C342
tower 3 down X102
Twr 3 Dwn Flt C362 RST
Twr 3 Lowering C342 RST
220
Lane 2 Go Down C431
2 Rows C20
3 Rows C21
Lane 3 Go Down C432
4 Rows C22
Twr 4 Lowering C343 SET
Co-Pro instruction may be used later date
SV4 Down S211 SET
221
Twr 4 Lowering C343
tower 4 down X103
TMR
Twr 4 Drop Time T203
Twr Down Max Tm V2162
Twr 4 Drop Time T203
Twr 4 Dwn Flt C363 SET
222
Twr 4 Lowering C343
tower 4 down X103
Twr 4 Dwn Flt C363 RST
Twr 4 Lowering C343 RST
223
Lane 2 Go Down C431
2 Rows C20
Lane 3 Go Down C432
3 Rows C21
4 Rows C22
Twr 5 Lowering C344 SET
Co-Pro instruction may be used later date
SV5 Down S220 SET

224
Twr 5 Lowering C344 — tower 5 down X104 — TMR Twr 5 Drop Time T204 Twr Down Max Tm V2162
Twr 5 Drop Time T204 — Twr 5 Dwn Flt C364 (SET)

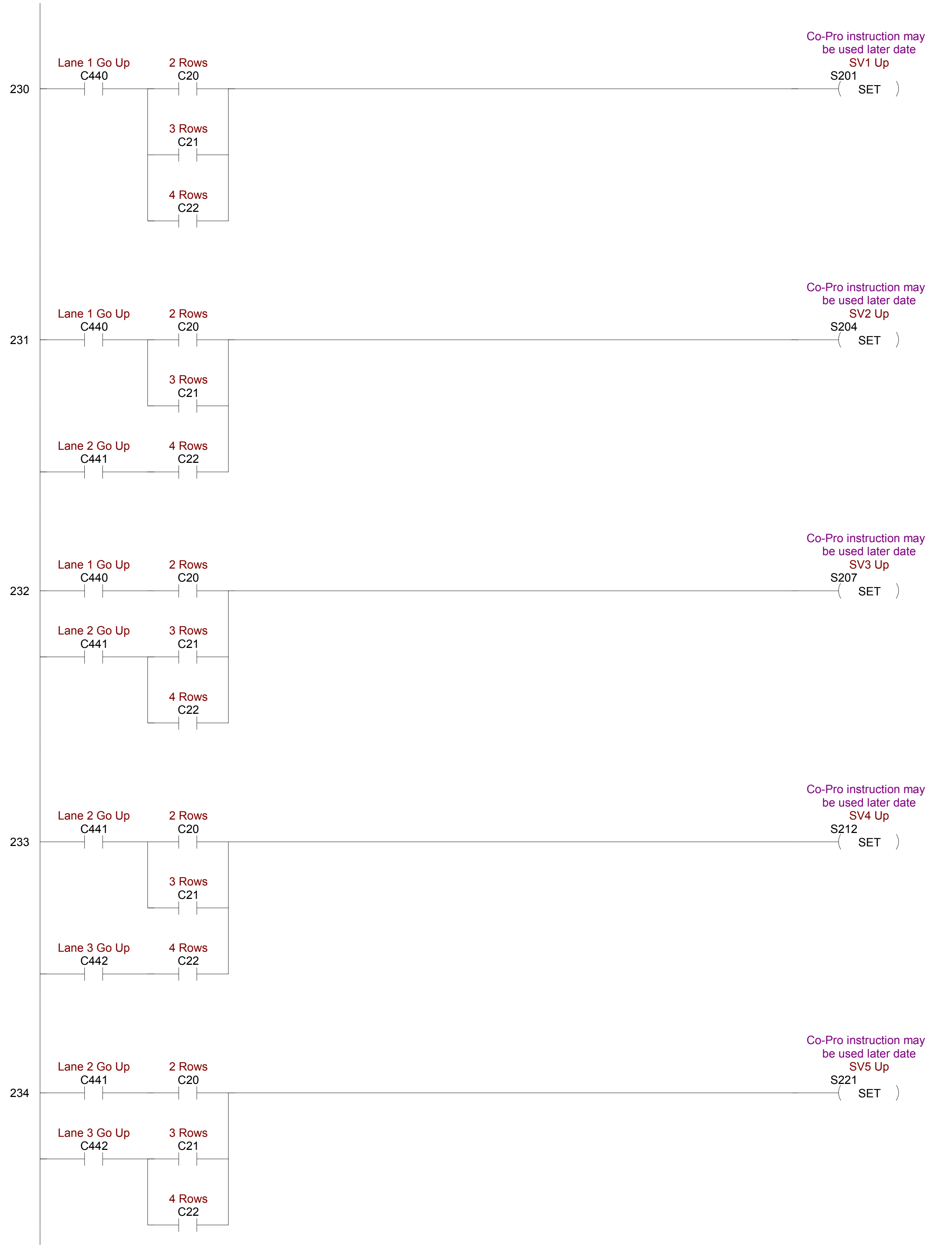
225
Twr 5 Lowering C344 — tower 5 down X104 — Twr 5 Lowering C344 (RST)
Twr 5 Dwn Flt C364 (RST)

226
Lane 2 Go Down C431 — 2 Rows C20 — Twr 6 Lowering C345 (SET)
Lane 3 Go Down C432 — 3 Rows C21
Lane 4 Go Down C433 — 4 Rows C22
Co-Pro instruction may be used later date
SV6 Down S223 (SET)

227
Twr 6 Lowering C345 — tower 6 down X105 — TMR Twr 6 Drop Time T205 Twr Down Max Tm V2162
Twr 6 Drop Time T205 — Twr 6 Dwn Flt C365 (SET)

228
Twr 6 Lowering C345 — tower 6 down X105 — Twr 6 Dwn Flt C365 (RST)
Twr 6 Lowering C345 (RST)

229
State Not Used _On SP1 — Lane 1 Go Down C430 C435 (RST)


230
Lane 1 Go Up
C440
2 Rows
C20
3 Rows
C21
4 Rows
C22
Co-Pro instruction may be used later date
SV1 Up
S201
SET
231
Lane 1 Go Up
C440
2 Rows
C20
3 Rows
C21
Lane 2 Go Up
C441
4 Rows
C22
Co-Pro instruction may be used later date
SV2 Up
S204
SET
232
Lane 1 Go Up
C440
2 Rows
C20
Lane 2 Go Up
C441
3 Rows
C21
4 Rows
C22
Co-Pro instruction may be used later date
SV3 Up
S207
SET
233
Lane 2 Go Up
C441
2 Rows
C20
3 Rows
C21
Lane 3 Go Up
C442
4 Rows
C22
Co-Pro instruction may be used later date
SV4 Up
S212
SET
234
Lane 2 Go Up
C441
2 Rows
C20
Lane 3 Go Up
C442
3 Rows
C21
4 Rows
C22
Co-Pro instruction may be used later date
SV5 Up
S221
SET

235
Lane 2 Go Up C441 — 2 Rows C20
Lane 3 Go Up C442 — 3 Rows C21
Lane 4 Go Up C443 — 4 Rows C22
Co-Pro instruction may be used later date SV6 Up
S224 (SET)

236
State Not Used _On SP1
Lane 1 Go Up C440 C445 (RST)

237
Lane 1 Do Step C450 — 2 Rows C20
3 Rows C21
4 Rows C22
Co-Pro instruction may be used later date SV1 Step
S202 (SET)

238
Lane 1 Do Step C450 — 2 Rows C20
3 Rows C21
Lane 2 Do Step C451 — 4 Rows C22
Co-Pro instruction may be used later date SV2 Step
S205 (SET)

239
Lane 1 Do Step C450 — 2 Rows C20
Lane 2 Do Step C451 — 3 Rows C21
4 Rows C22
Co-Pro instruction may be used later date SV3 Step
S210 (SET)

240
Lane 2 Do Step C451 — 2 Rows C20 / 3 Rows C21
Lane 3 Do Step C452 — 4 Rows C22
Co-Pro instruction may be used later date SV4 Step S213 (SET)

241
Lane 2 Do Step C451 — 2 Rows C20
Lane 3 Do Step C452 — 3 Rows C21 / 4 Rows C22
Co-Pro instruction may be used later date SV5 Step S222 (SET)

242
Lane 2 Do Step C451 — 2 Rows C20
Lane 3 Do Step C452 — 3 Rows C21
Lane 4 Do Step C453 — 4 Rows C22
Co-Pro instruction may be used later date SV6 Step S225 (SET)

243
State Not Used _On SP1
Lane 1 Do Step C450 C455 (RST)

SG
Co-Pro instruction may be used later date SV1 Down
S200

K1 = Tower A Down, K2 = Tower A Step, K3 = Tower A Up,

245
Co-Pro instruction may be used later date SV1 Down S200 (RST)
Drive 1 Trigger Y100 (RST)
Move Tower 1 C260 (SET)

SG
Co-Pro instruction may be used later date
SV1 Up
S201

247
Move Tower 1
C260

Drive 1 Trigger
Y100
SET

Co-Pro instruction may be used later date
SV1 Up
S201
RST

Move Tower 1
C260
RST

SG
Co-Pro instruction may be used later date
SV1 Step
S202

249
TMR
Step 1 Time
T121
K2

250
Drive 1 @ Target
X10
Move Tower 1
C260

Drive 1 Step
Y101
OUT

251
Step 1 Time
T121

Co-Pro instruction may be used later date
SV1 Step
S202
RST

SG
Co-Pro instruction may be used later date
SV2 Down
S203

k4 = Tower B Down, k5 = Tower B Step, k6 = Tower B Up
K1 = Tower A Down, K2 = Tower A Step, K3 = Tower A Up,

253
Drive 2 Trigger
Y104
RST

Co-Pro instruction may be used later date
SV2 Down
S203
RST

Move Tower 2
C261
SET

SG
Co-Pro instruction may be used later date
SV2 Up
S204

255
Move Tower 2
C261

Drive 2 Trigger
Y104
SET

Co-Pro instruction may be used later date
SV2 Up
S204
RST

Move Tower 2
C261
RST

SG
Co-Pro instruction may be used later date
SV2 Step
S205

257

TMR
Step 2 Time
T122
K2

258
Drive 2 @ Target
X12

Move Tower 2
C261

Drive 2 Step
Y105
OUT

259
Step 2 Time
T122

Co-Pro instruction may be used later date
SV2 Step
S205
RST

SG
Co-Pro instruction may be used later date
SV3 Down
S206

k7 = Tower C Down, k8 = Tower C Step, k9 = Tower C Up

K1 = Tower A Down, K2 = Tower A Step, K3 = Tower A Up,

261

Co-Pro instruction may be used later date
SV3 Down
S206
RST

Drive 3 Trigger
Y106
RST

Move Tower 3
C262
SET

SG
Co-Pro instruction may be used later date
SV3 Up
S207

263
Move Tower 3
C262

Drive 3 Trigger
Y106
SET

Co-Pro instruction may be used later date
SV3 Up
S207
RST

Move Tower 3
C262
RST

SG
Co-Pro instruction may be used later date
SV3 Step
S210

265
TMR
Step 3 Time
T123
K2

266
Drive 3 @ Target
X14

Move Tower 3
C262

Drive 3 Step
Y107
OUT

267
Step 3 Time
T123

Co-Pro instruction may be used later date
SV3 Step
S210
RST

SG
Co-Pro instruction may be used later date
SV4 Down
S211

ka = Tower D Down, kb = Tower D Step, kc = Tower D Up
K1 = Tower A Down, K2 = Tower A Step, K3 = Tower A Up,

269
Co-Pro instruction may be used later date
SV4 Down
S211
RST

Drive 4 Trigger
Y110
RST

Move Tower 4
C263
SET

SG
Co-Pro instruction may be used later date
SV4 Up
S212

271
Move Tower 4
C263

Drive 4 Trigger
Y110
SET

Co-Pro instruction may be used later date
SV4 Up
S212
RST

Move Tower 4
C263
RST

SG
Co-Pro instruction may be used later date
SV4 Step
S213

273
TMR
Step 4 Time
T124
K2

274
Drive 4 @ Target
X16

Move Tower 4
C263

Drive 4 Step
Y111
OUT

275
Step 4 Time
T124

Co-Pro instruction may be used later date
SV4 Step
S213
RST

SG
Co-Pro instruction may be used later date
SV5 Down
S220

K1 = Tower A Down, K2 = Tower A Step, K3 = Tower A Up,

K1 = Tower A Down, K2 = Tower A Step, K3 = Tower A Up,

277
Co-Pro instruction may be used later date
SV5 Down
S220
RST

Drive 5 Trigger
Y112
RST

Move Tower 5
C264
SET

SG
Co-Pro instruction may be used later date
SV5 Up
S221

279
Move Tower 5
C264
Drive 5 Trigger
Y112
SET
Co-Pro instruction may be used later date
SV5 Up
S221
RST
Move Tower 5
C264
RST

SG
Co-Pro instruction may be used later date
SV5 Step
S222

281
TMR
Step 5 Time
T125
K2

282
Drive 5 @ Target
X106
Move Tower 5
C264
Drive 5 Step
Y113
OUT

283
Step 5 Time
T125
Co-Pro instruction may be used later date
SV5 Step
S222
RST

SG
Co-Pro instruction may be used later date
SV6 Down
S223

k4 = Tower B Down, k5 = Tower B Step, k6 = Tower B Up
K1 = Tower A Down, K2 = Tower A Step, K3 = Tower A Up,

285
Co-Pro instruction may be used later date
SV6 Down
S223
RST
Drive 6 Trigger
Y114
RST
Move Tower 6
C265
SET

SG
Co-Pro instruction may be used later date
SV6 Up
S224

287
Move Tower 6
C265

Drive 6 Trigger
Y114
(SET)

Co-Pro instruction may be used later date
SV6 Up
S224
(RST)

Move Tower 6
C265
(RST)

SG
Co-Pro instruction may be used later date
SV6 Step
S225

289
TMR
Step 6 Time
T126
K2

290
Drive 6 @ Target
X110

Move Tower 6
C265

Drive 6 Step
Y115
(OUT)

291
Step 6 Time
T126

Co-Pro instruction may be used later date
SV6 Step
S225
(RST)

SG
SV7 Down
S226

k7 = Tower C Down, k8 = Tower C Step, k9 = Tower C Up

K1 = Tower A Down, K2 = Tower A Step, K3 = Tower A Up,

293

Move Tower 7
C266

LDD
K0

OUTD
Lane 7 Position
V2134

LDD
From Display
Dn Vel
V2442

OUTD
Vel Lan 7
V2270

LDD
From Display
Dn Acel
V2446

OUTD
Accel Lan 7
V2272

LDD
From Display
Dn Decel
V2452

OUTD
Decel Lan7
V2274

Move Tower 7
C266
(SET)

SV7 Down
S226
(RST)

SG
SV7 Up
S227

295

Move Tower 7
C266

LDD
In mm
From Display
Linmot Up Dist
V4040

OUTD
Lane 7 Position
V2134

LDD
From Display
Uo Vel
V2440

OUTD
Vel Lan 7
V2270

LDD
From Display
Up Acel
V2444

OUTD
Accel Lan 7
V2272

LDD
From Display
Up Decel
V2450

OUTD
Decel Lan7
V2274

Move Tower 7
C266
(SET)

SV7 Up
S227
(RST)

SG
SV7 Step
S230

297

Lane 7 Position
V2134 ≠ K0

Move Tower 7
C266

LDD
Lane 7 Position
V2134

SUBBD
In mm
From Display
Linmot Step Dist
V4042

OUTD
Lane 7 Position
V2134

LDD
From Display
Uo Vel
V2440

OUTD
Vel Lan 7
V2270

LDD
From Display
Up Acel
V2444

OUTD
Accel Lan 7
V2272

LDD
From Display
Up Decel
V2450

OUTD
Decel Lan7
V2274

Move Tower 7
C266
(SET)

SV7 Step
S230
(RST)

SG
SV8 Down
S231

ka = Tower D Down, kb = Tower D Step, kc = Tower D Up
k7 = Tower C Down, k8 = Tower C Step, k9 = Tower C Up

K1 = Tower A Down, K2 = Tower A Step, K3 = Tower A Up,

299

Move Tower 8
C267

LDD
K0

OUTD
Lane 8 Position
V2136

LDD
From Display
Dn Vel
V2442

OUTD
Vel Lan 8
V2300

LDD
From Display
Dn Acel
V2446

OUTD
Accel Lan 8
V2302

LDD
From Display
Dn Decel
V2452

OUTD
Decel Lan8
V2304

Move Tower 8
C267
(SET)

SV8 Down
S231
(RST)

SG
SV8 Up
S232

301

Move Tower 8
C267

LDD
In mm
From Display
Linmot Up Dist
V4040

OUTD
Lane 8 Position
V2136

LDD
From Display
Uo Vel
V2440

OUTD
Vel Lan 8
V2300

LDD
From Display
Up Acel
V2444

OUTD
Accel Lan 8
V2302

LDD
From Display
Up Decel
V2450

OUTD
Decel Lan8
V2304

Move Tower 8
C267
(SET)

SV8 Up
S232
(RST)

SG
SV8 Step
S233

303
Lane 8 Position V2136 ≠ K0 — Move Tower 8 C267 (normally closed)

LDD Lane 8 Position V2136
SUBBD In mm From Display Linmot Step Dist V4042
OUTD Lane 8 Position V2136
LDD From Display Uo Vel V2440
OUTD Vel Lan 8 V2300
LDD From Display Up Acel V2444
OUTD Accel Lan 8 V2302
LDD From Display Up Decel V2450
OUTD Decel Lan8 V2304
Move Tower 8 C267 (SET)
SV8 Step S233 (RST)

This timer done instead of testing the second section of the real time clock because
the RTC isn't accurate if not initialized.

SG
Rate Accum
S300

305
One Minute Tmr T32 (normally closed)

TMR One Minute Tmr T32 K600

306
One Minute Tmr T32

Minute One Shot C0 (PD)

307

Minute One Shot
C0

LD
Lane 1 Rate
CTA10

ADD
Lane 2 Rate
CTA11

ADD
Lane 3 Rate
CTA12

ADD
Lane 4 Rate
CTA13

MUL
K5

OUT
All Lanes Rate
V2044

Rate Lane 1
CT10 CT16
(RST)

SG
Total Accum
S310

309

LDD
Lane 1 Accum
V2020

ADDD
Lane 2 Accum
V2022

ADDD
Lane 3 Accum
V2024

ADDD
Lane 4 Accum
V2026

OUTD
All Lanes Accum
V2045

310
Accum Zero
C103

LDD
K0

OUTD
Lane 1 Accum
V2020

OUTD
Lane 2 Accum
V2022

OUTD
Lane 3 Accum
V2024

OUTD
Lane 4 Accum
V2026

SG
Misc Control
S340

312
From HMI
Clear Btn Row 1
C140

Row 1 PE Blocked
C330

Run
C4

Guard Doors OK
X21

Row 1 Clearing
C200
(SET)

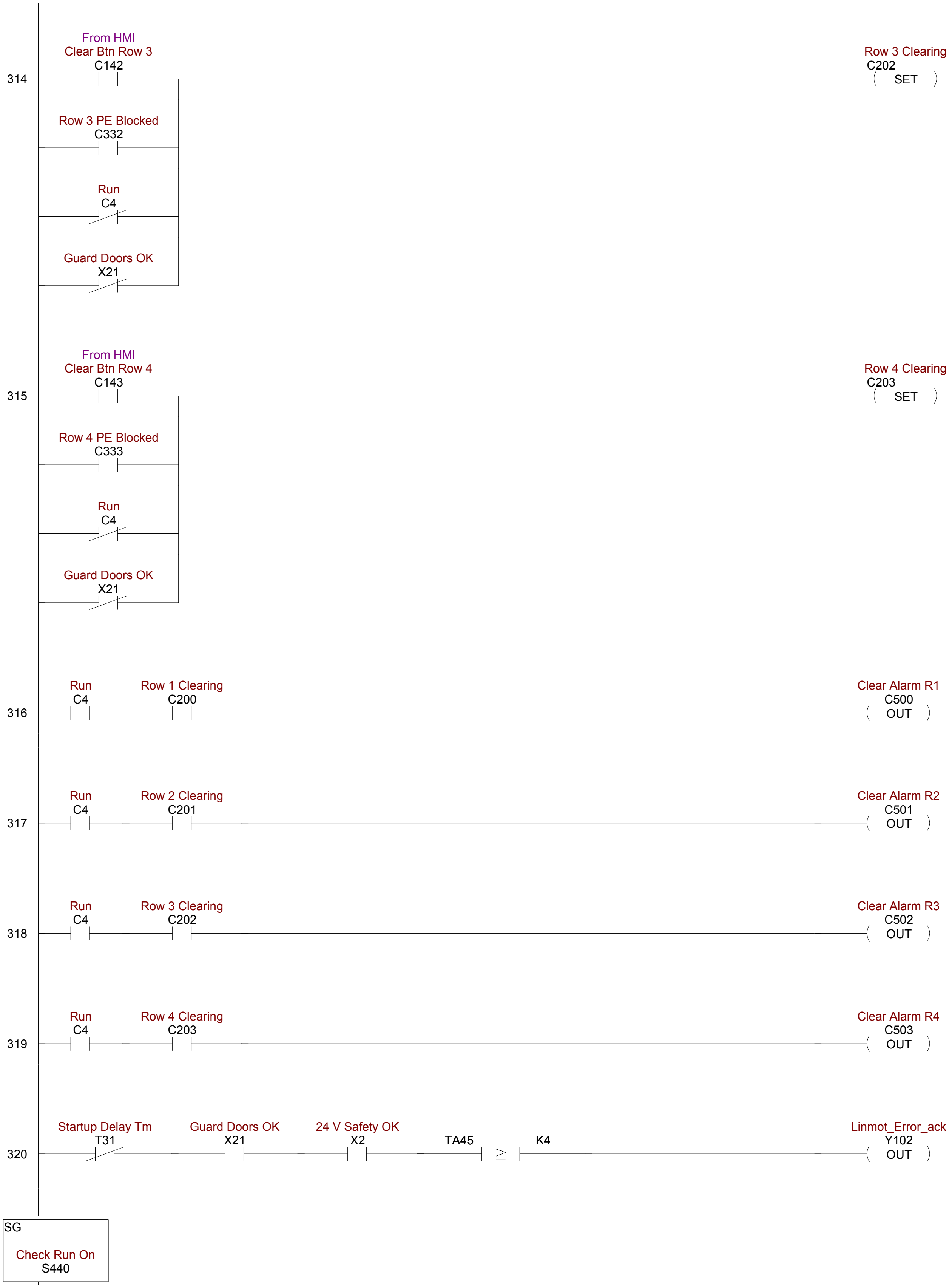
313
From HMI
Clear Btn Row 2
C141

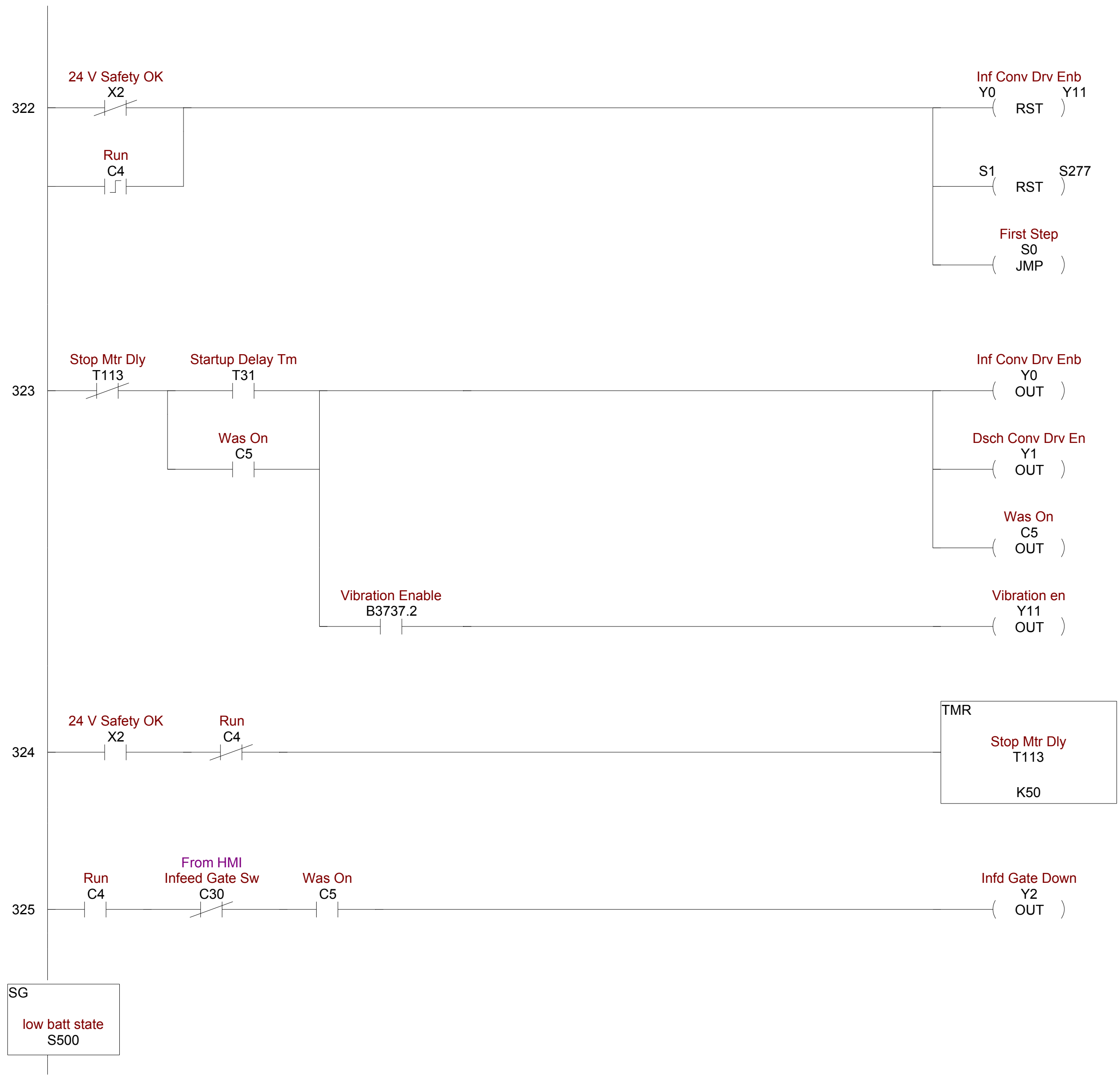
Row 2 PE Blocked
C331

Run
C4

Guard Doors OK
X21

Row 2 Clearing
C201
(SET)


314
From HMI
Clear Btn Row 3
C142
Row 3 PE Blocked
C332
Run
C4
Guard Doors OK
X21
Row 3 Clearing
C202
SET
315
From HMI
Clear Btn Row 4
C143
Row 4 PE Blocked
C333
Run
C4
Guard Doors OK
X21
Row 4 Clearing
C203
SET
316
Run
C4
Row 1 Clearing
C200
Clear Alarm R1
C500
OUT
317
Run
C4
Row 2 Clearing
C201
Clear Alarm R2
C501
OUT
318
Run
C4
Row 3 Clearing
C202
Clear Alarm R3
C502
OUT
319
Run
C4
Row 4 Clearing
C203
Clear Alarm R4
C503
OUT
320
Startup Delay Tm
T31
Guard Doors OK
X21
24 V Safety OK
X2
TA45
≥
K4
Linmot_Error_ack
Y102
OUT
SG
Check Run On
S440


322
24 V Safety OK
X2
Run
C4
Inf Conv Drv Enb
Y0
Y11
RST
S1
S277
RST
First Step
S0
JMP
323
Stop Mtr Dly
T113
Startup Delay Tm
T31
Was On
C5
Inf Conv Drv Enb
Y0
OUT
Dsch Conv Drv En
Y1
OUT
Was On
C5
OUT
Vibration Enable
B3737.2
Vibration en
Y11
OUT
324
24 V Safety OK
X2
Run
C4
TMR
Stop Mtr Dly
T113
K50
325
Run
C4
From HMI
Infeed Gate Sw
C30
Was On
C5
Infd Gate Down
Y2
OUT
SG
low batt state
S500

327
ack. low batt
C53

LD
prev page
V2061

OUT
Page Request
V2055

LD
K999

OUT
prev page
V2061

low batt state
S500
RST

ack. low batt
C53
RST

SG
For HMI Machine Status Window Messages
S760

329
current page
V2050 = K500

Splash Screen Tm
T371

TMR
Splash Screen Tm
T371
K80

330
Splash Screen Tm
T371

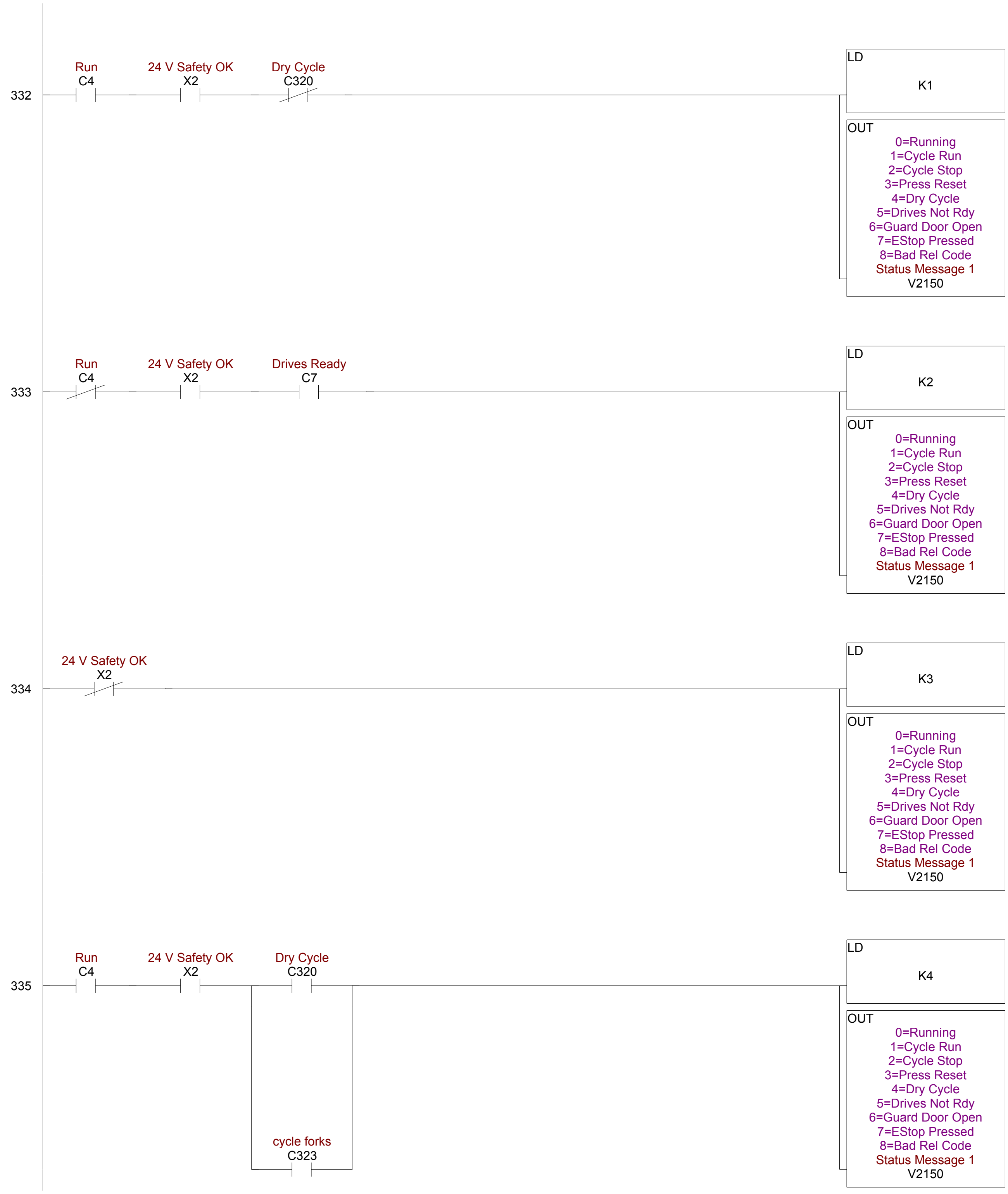
LD
K1

OUT
Page Request
V2055

331
State Not Used
_On
SP1

LD
K0

OUT
0=Running
1=Cycle Run
2=Cycle Stop
3=Press Reset
4=Dry Cycle
5=Drives Not Rdy
6=Guard Door Open
7=EStop Pressed
8=Bad Rel Code
Status Message 1
V2150


332
Run
C4
24 V Safety OK
X2
Dry Cycle
C320
LD
K1
OUT
0=Running
1=Cycle Run
2=Cycle Stop
3=Press Reset
4=Dry Cycle
5=Drives Not Rdy
6=Guard Door Open
7=EStop Pressed
8=Bad Rel Code
Status Message 1
V2150
333
Run
C4
24 V Safety OK
X2
Drives Ready
C7
LD
K2
OUT
0=Running
1=Cycle Run
2=Cycle Stop
3=Press Reset
4=Dry Cycle
5=Drives Not Rdy
6=Guard Door Open
7=EStop Pressed
8=Bad Rel Code
Status Message 1
V2150
334
24 V Safety OK
X2
LD
K3
OUT
0=Running
1=Cycle Run
2=Cycle Stop
3=Press Reset
4=Dry Cycle
5=Drives Not Rdy
6=Guard Door Open
7=EStop Pressed
8=Bad Rel Code
Status Message 1
V2150
335
Run
C4
24 V Safety OK
X2
Dry Cycle
C320
cycle forks
C323
LD
K4
OUT
0=Running
1=Cycle Run
2=Cycle Stop
3=Press Reset
4=Dry Cycle
5=Drives Not Rdy
6=Guard Door Open
7=EStop Pressed
8=Bad Rel Code
Status Message 1
V2150

336
Run C4 — 24 V Safety OK X2 — Drives Ready C7

LD K5

OUT
0=Running
1=Cycle Run
2=Cycle Stop
3=Press Reset
4=Dry Cycle
5=Drives Not Rdy
6=Guard Door Open
7=EStop Pressed
8=Bad Rel Code
Status Message 1
V2150

337
Guard Doors OK X21 — 24 V Safety OK X2

LD K6

OUT
0=Running
1=Cycle Run
2=Cycle Stop
3=Press Reset
4=Dry Cycle
5=Drives Not Rdy
6=Guard Door Open
7=EStop Pressed
8=Bad Rel Code
Status Message 1
V2150

338
EStops OK X20

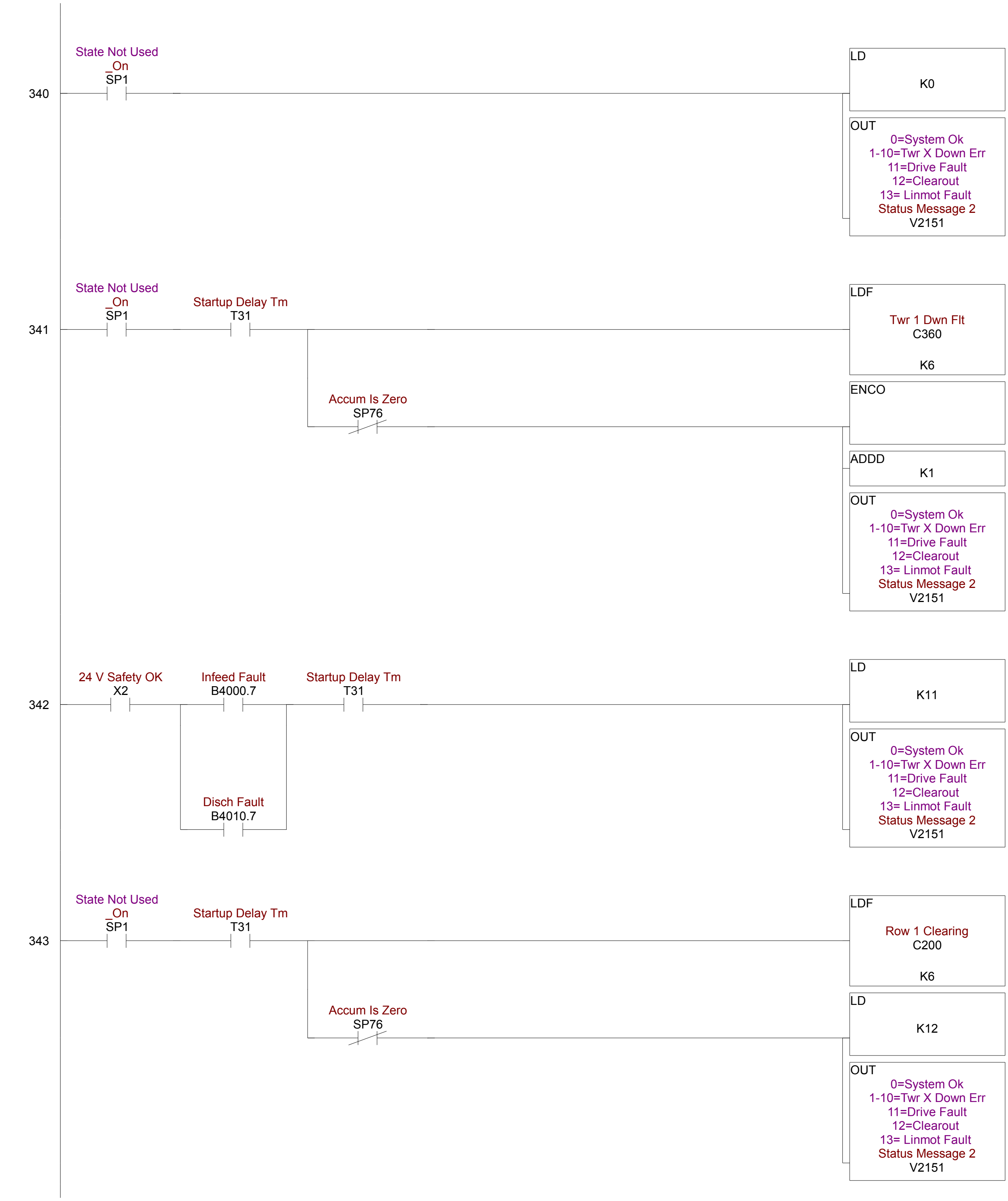
LD K7

OUT
0=Running
1=Cycle Run
2=Cycle Stop
3=Press Reset
4=Dry Cycle
5=Drives Not Rdy
6=Guard Door Open
7=EStop Pressed
8=Bad Rel Code
Status Message 1
V2150

339
Code Not Entered C325

LD K8

OUT
0=Running
1=Cycle Run
2=Cycle Stop
3=Press Reset
4=Dry Cycle
5=Drives Not Rdy
6=Guard Door Open
7=EStop Pressed
8=Bad Rel Code
Status Message 1
V2150


340
State Not Used
_On
SP1
LD
K0
OUT
0=System Ok
1-10=Twr X Down Err
11=Drive Fault
12=Clearout
13= Linmot Fault
Status Message 2
V2151
341
State Not Used
_On
SP1
Startup Delay Tm
T31
Accum Is Zero
SP76
LDF
Twr 1 Dwn Flt
C360
K6
ENCO
ADDD
K1
OUT
0=System Ok
1-10=Twr X Down Err
11=Drive Fault
12=Clearout
13= Linmot Fault
Status Message 2
V2151
342
24 V Safety OK
X2
Infeed Fault
B4000.7
Disch Fault
B4010.7
Startup Delay Tm
T31
LD
K11
OUT
0=System Ok
1-10=Twr X Down Err
11=Drive Fault
12=Clearout
13= Linmot Fault
Status Message 2
V2151
343
State Not Used
_On
SP1
Startup Delay Tm
T31
Accum Is Zero
SP76
LDF
Row 1 Clearing
C200
K6
LD
K12
OUT
0=System Ok
1-10=Twr X Down Err
11=Drive Fault
12=Clearout
13= Linmot Fault
Status Message 2
V2151

344

24 V Safety OK X2 — Drive 1 Faulted X11 / Drive 2 Faulted X13 / Drive 3 Faulted X15 / Drive 4 Faulted X17 / Drive 5 Faulted X107 / Drive 6 Faulted X111

LD K13

OUT
0=System Ok
1-10=Twr X Down Err
11=Drive Fault
12=Clearout
13= Linmot Fault
Status Message 2
V2151

Linmot Faulted
C32
OUT

Linmots homed
C31
RST

SG
Port Setup 1
S770

Setup Secondary Comm Port for Modbus (K20 -> V7655), No Parity, 2 Stop Bits, RS485 Echo Supression, 19200 Baud, Local Address 1 (K2e01 -> V7656). Then initiate change by (K500 -> V7657)

346

LD K20

OUT
Port Set Reg 1
V7655

LD K2e01

OUT
Port Set Reg 2
V7656

LD K500

OUT
Port Set Reg 3
V7657

Port Setup 2
S771
JMP

SG
Port Setup 2
S771

Wait for port setup acknowledge by (Ka00 -> V7657)

348

Port Set Reg 3
V7657 = Ka00

Set Infeed Speed
S1000
JMP

SG
Set Infeed Speed
S1000

350
Port 2 Busy
SP116

LD
Infeed Speed
V3726

BIN

OUT
Decimal Speed
V3600

MWX

CPU/DCM Slot :	CPU
Port Number :	K2
Slave Address :	K100
Function Code :	06 - Preset Single Register
Start Slave Memory Address :	K48194
Start Master Memory Address :	V3600
Number of Elements :	n/a
Modbus Data type :	584/984 Mode
Exception Response Buffer :	V400

Read Infeed Stat
S1001
JMP

SG
Read Infeed Stat
S1001

352
Port 2 Busy
SP116

MRX

CPU/DCM Slot :	CPU
Port Number :	K2
Slave Address :	K100
Function Code :	03 - Read Holding Registers
Start Slave Memory Address :	K48449
Start Master Memory Address :	V4000
Number of Elements :	K7
Modbus Data type :	584/984 Mode
Exception Response Buffer :	V400

Read Infeed Temp
S1002
JMP

SG
Read Infeed Temp
S1002

354
Port 2 Busy
SP116

MRX

CPU/DCM Slot :	CPU
Port Number :	K2
Slave Address :	K100
Function Code :	03 - Read Holding Registers
Start Slave Memory Address :	K40028
Start Master Memory Address :	V4007
Number of Elements :	K1
Modbus Data type :	584/984 Mode
Exception Response Buffer :	V400

Read DC Ripple
S1003
JMP

SG
Read DC Ripple
S1003

356
Port 2 Busy
SP116

MRX
CPU/DCM Slot : CPU
Port Number : K2
Slave Address : K100
Function Code : 03 - Read Holding Registers
Start Slave Memory Address : K40381
Start Master Memory Address : V4101
Number of Elements : K1
Modbus Data type : 584/984 Mode
Exception Response Buffer : V400

Set Dsch Speed
S1010
JMP

SG
Set Dsch Speed
S1010

358
Port 2 Busy
SP116

LD
Vibration Speed
V3725

BIN

OUT
Decimal Speed
V3600

MWX
CPU/DCM Slot : CPU
Port Number : K2
Slave Address : K101
Function Code : 06 - Preset Single Register
Start Slave Memory Address : K48194
Start Master Memory Address : V3600
Number of Elements : n/a
Modbus Data type : 584/984 Mode
Exception Response Buffer : V400

Read Disch Stat
S1011
JMP

SG
Read Disch Stat
S1011

360
Port 2 Busy
SP116

MRX
CPU/DCM Slot : CPU
Port Number : K2
Slave Address : K101
Function Code : 03 - Read Holding Registers
Start Slave Memory Address : K48449
Start Master Memory Address : V4010
Number of Elements : K7
Modbus Data type : 584/984 Mode
Exception Response Buffer : V400

Read Disch Temp
S1012
JMP

SG
Read Disch Temp
S1012

362
Port 2 Busy
SP116

MRX

CPU/DCM Slot :	CPU
Port Number :	K2
Slave Address :	K101
Function Code :	03 - Read Holding Registers
Start Slave Memory Address :	K40028
Start Master Memory Address :	V4017
Number of Elements :	K1
Modbus Data type :	584/984 Mode
Exception Response Buffer :	V400

Read Disch Rippl
S1017
JMP

SG
Read Disch Rippl
S1017

364
Port 2 Busy
SP116

MRX

CPU/DCM Slot :	CPU
Port Number :	K2
Slave Address :	K101
Function Code :	03 - Read Holding Registers
Start Slave Memory Address :	K40381
Start Master Memory Address :	V4102
Number of Elements :	K1
Modbus Data type :	584/984 Mode
Exception Response Buffer :	V400

Set Vib Speed
S1013
JMP

SG
Set Vib Speed
S1013

366
Port 2 Busy
SP116

LD
Discharge Speed
V3727

BIN

OUT
Decimal Speed
V3600

MWX

CPU/DCM Slot :	CPU
Port Number :	K2
Slave Address :	K102
Function Code :	06 - Preset Single Register
Start Slave Memory Address :	K48194
Start Master Memory Address :	V3600
Number of Elements :	n/a
Modbus Data type :	584/984 Mode
Exception Response Buffer :	V400

Read Vib Stat
S1014
JMP

SG
Read Vib Stat
S1014

368 Port 2 Busy SP116

MRX
CPU/DCM Slot : CPU
Port Number : K2
Slave Address : K102
Function Code : 03 - Read Holding Registers
Start Slave Memory Address : K48449
Start Master Memory Address : V4020
Number of Elements : K7
Modbus Data type : 584/984 Mode
Exception Response Buffer : V400

Read Vib Temp
S1015
JMP

SG
Read Vib Temp
S1015

370 Port 2 Busy SP116

MRX
CPU/DCM Slot : CPU
Port Number : K2
Slave Address : K102
Function Code : 03 - Read Holding Registers
Start Slave Memory Address : K40028
Start Master Memory Address : V4027
Number of Elements : K1
Modbus Data type : 584/984 Mode
Exception Response Buffer : V400

Read Vib Ripple
S1016
JMP

SG
Read Vib Ripple
S1016

372 Port 2 Busy SP116

MRX
CPU/DCM Slot : CPU
Port Number : K2
Slave Address : K102
Function Code : 03 - Read Holding Registers
Start Slave Memory Address : K40381
Start Master Memory Address : V4103
Number of Elements : K1
Modbus Data type : 584/984 Mode
Exception Response Buffer : V400

Set Infeed Speed
S1000
JMP

SG
Check for valid login
from HMI
HMI Login
S1770

Compare the password entered from the LOGIN button on the HMI against the stored passwords. There are three access levels...Shift_ Oper PSW is the lowest, Shift_Maint PSW is the second level, and Admin PSW (constant value 7734 - not user definable) is the highest. The appropriate PSW Level _ bits are set based on the current login level to control access of specific HMI buttons via visibility tags.

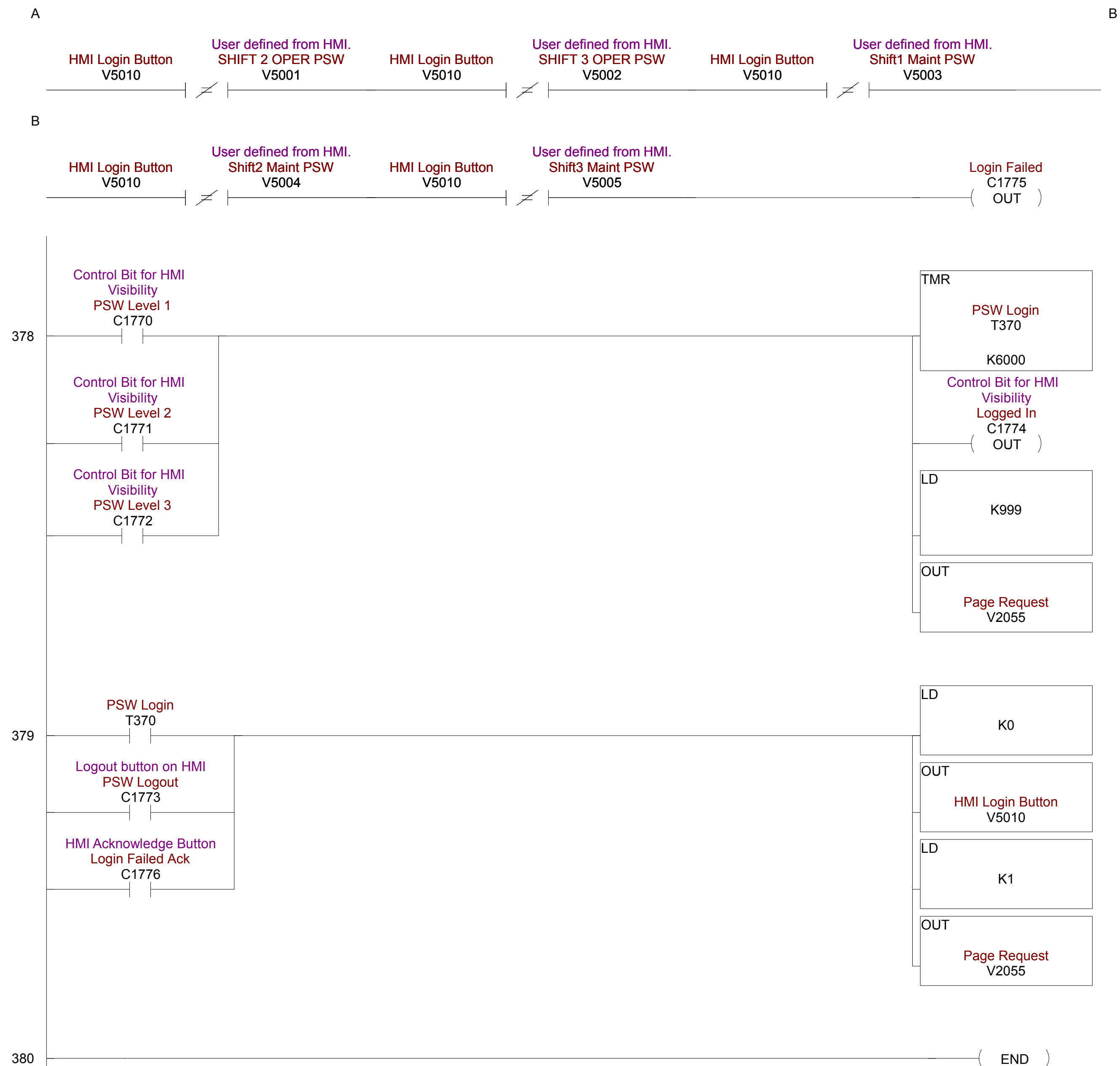
374
HMI Login Button
V5010 = K7734
HMI Login Button
V5010 = K8324
LD
K3
OUT
K1=Level 1
K2=Level 2
K3=Level 3
PSW Level
V5012
Control Bit for HMI Visibility
PSW Level 1
C1770
OROUT
Control Bit for HMI Visibility
PSW Level 2
C1771
OROUT
Control Bit for HMI Visibility
PSW Level 3
C1772
OROUT

375
HMI Login Button
V5010 = User defined from HMI. Shift1 Maint PSW V5003
HMI Login Button
V5010 = User defined from HMI. Shift2 Maint PSW V5004
HMI Login Button
V5010 = User defined from HMI. Shift3 Maint PSW V5005
LD
K2
OUT
K1=Level 1
K2=Level 2
K3=Level 3
PSW Level
V5012
Control Bit for HMI Visibility
PSW Level 1
C1770
OROUT
Control Bit for HMI Visibility
PSW Level 2
C1771
OROUT

376
HMI Login Button
V5010 = User defined from HMI. SHIFT 1 OPER PSW V5000
HMI Login Button
V5010 = User defined from HMI. SHIFT 2 OPER PSW V5001
HMI Login Button
V5010 = User defined from HMI. SHIFT 3 OPER PSW V5002
LD
K1
OUT
K1=Level 1
K2=Level 2
K3=Level 3
PSW Level
V5012
Control Bit for HMI Visibility
PSW Level 1
C1770
OROUT

A

377
HMI Login Button V5010 ≠ K0
HMI Login Button V5010 ≠ K7734
HMI Login Button V5010 ≠ K8324
HMI Login Button V5010 ≠ User defined from HMI. SHIFT 1 OPER PSW V5000



This subroutine inspects the entered release codes. If they match this month's release code or the permanent release code then it returns with C400 SET else C400 is RST

SBR
K1

Load the release code table into RAM memory. RST the 'Correct Code' bit. Using the current year and month select the current month's release code from the table.

382

LD K79

LD K0

LDLBL K2

MOVMC ((Year * 12) + month) Year-Month V2310

Correct Code C324 (RST)

LD Current Year V7774

MUL K12

ADD Current Month V7773

SUB ((Year * 12) + month) Year-Month V2310

MUL K2

BIN

ADDB K4c3

OUT First Ptr V2510

LDD P2510

OUTD Double Month Rel Code V2512

Compare the two displayed release codes with the current month's release code and with the permanent release code. If any match then SET C400 Note: there are two displayed release codes so they can enter the next month's entry and not have to worry about changing right at the month changeover.

383

State Not Used _On SP1

LDD Month Entry 1-1 V2500

BIN

CMPD Double Month Rel Code V2512

Is Equal SP61

Correct Code C324 (SET)

384
Correct Code
C324
LDD
Month Entry 1-1
V2500
BIN
CMPD
Perm Rel
V2311
Is Equal
SP61
Correct Code
C324
SET

385
Correct Code
C324
LDD
Month Entry 2-1
V2502
BIN
CMPD
Double
Month Rel Code
V2512
Is Equal
SP61
Correct Code
C324
SET

386
Correct Code
C324
LDD
Month Entry 2-1
V2502
BIN
CMPD
Perm Rel
V2311
Is Equal
SP61
Correct Code
C324
SET

Finally zero out the RAM image of the release codes. They should never be visible to a casual observer. For testing add an NC of SP1 in series to disable the rung.

387
State Not Used
_On
SP1
LD
K7b
LDA
O2300
FILL
K0

388
RT

Process the diameter information

SBR
K2

390

LD K2

OUT Min Rows V2100

LD K5

OUT Max Rows V2101

391

Row Count V3732 < Min Rows V2100

LD Min Rows V2100

OUT Row Count V3732

392

Row Count V3732 ≥ Max Rows V2101

LD Max Rows V2101

OUT Row Count V3732

Once the tortillia size is checked for validity, use it as a pointer into the table to generate the travel and down time pointers.

393

State Not Used _On SP1

LD Row Count V3732

SUB Min Rows V2100

MUL K2

OUT Row Offset V2104

LDA O2200

BCD

ADD Row Offset V2104

BIN

OUT Travel Time Ptr V2004

OUT Down Time Ptr V2005

INCB Down Time Ptr V2005

394 —(RT)

SBR K9

Moves 32 words from place pointed to by 'Case Index' to central storage starting at V3700

396

- LD Case Current V4050
- MUL K64
- BIN
- OUT Case Offset V4051
- LDA O10000
- ADDB Case Offset V4051
- OUT Store Case Add V4053
- LD K40
- LD Store Case Add V4053
- MOV Start 64 Recipe Words Product Name V3700

397 State Not Used _On SP1 — Lower Lower Fork Cnt V3730 ≥ Stack Cnt Target V3720

- LD Stack Cnt Target V3720
- SUBD K1
- OUT Lower Lower Fork Cnt V3730

398 Step Down Count V3731 = K0

- LD K1
- OUT Step Down Count V3731

399 State Not Used _On SP1 — 2 Rows C20 C27 (RST)

400 Row Count V3732 = K2 — 2 Rows C20 (SET)

401 Row Count V3732 = K3 — 3 Rows C21 (SET)

402 Row Count V3732 = K4 — 4 Rows C22 (SET)

403 (RT)

SBR K10

Moves 24 words from Central case storage (starting at V3700) to the storage pointed to by 'Case Index'

405 State Not Used _On SP1

LD	Case Current V4050
MUL	K64
BIN	
OUT	Case Offset V4051
LDA	O10000
ADDB	Case Offset V4051
OUT	Store Case Add V4053
LD	K40
LDA	O3700
MOV	Store Case Ptr P4053

406 (RT)

Following ar the eight time constants. These are travel and down time for each of 2 through 5 rows. The order is: 2 row travel, 2 row down, 3 row travel ...

DLBL K1

408 NCON K29

409 NCON K130

410 NCON K27

411 NCON K130

412 NCON K25

413 NCON K130

414 NCON K23

415 NCON K130

The following is ((2 * Number of Months) + 3) entries long. The first is year/month representation of the start date of the table.. The second 2 are the permanent release code. Be sure to comment the beginning year/month.

DLBL K2

This is the starting year (eg 05) * 12 plus the starting month (January = 1)

417 NCON K91

These two entries form the permanent release code. All the release entries are the hex equivalent of the number the operator will be typing into the display. This is one more small hurdle to block stealing the release codes.

418 NCON K99dd

419 NCON K1bc

First of the 'shipped' codes

420 NCON Kb0cf

421 NCON K309

Second of the 'shipped' codes

422 NCON Kd2ed

423	NCON	K1c4
424	NCON	Kf50b
425	NCON	K7f
426	NCON	Kf829
427	NCON	K530
428	NCON	K1a46
429	NCON	K3ec
430	NCON	K3c64
431	NCON	K2a7
432	NCON	Kbbfb
433	NCON	K77
434	NCON	K77f7
435	NCON	Kef
436	NCON	K33f2
437	NCON	K167

438	NCON	Kefee
439	NCON	K1de
440	NCON	Kabea
441	NCON	K256
442	NCON	K67e5
443	NCON	K2ce
444	NCON	K23e1
445	NCON	K346

End of 12 months codes

446	NCON	Kdfdd
447	NCON	K3bd
448	NCON	K9bd8
449	NCON	K435
450	NCON	K57d4
451	NCON	K4ad
452	NCON	K13d0
453	NCON	K525

454	NCON	Kcfcb
455	NCON	K59c
456	NCON	K55d9
457	NCON	K234
458	NCON	Kabb3
459	NCON	K468
460	NCON	K208c
461	NCON	Ka7
462	NCON	K7666
463	NCON	K2db
464	NCON	Kcc40
465	NCON	K50f
466	NCON	K4119
467	NCON	K14e
468	NCON	K96f3

469 NCON K382

End 24 months entry

470 NCON Keccc

471 NCON K5b6

472 NCON K61a6

473 NCON K1f5

474 NCON Kb780

475 NCON K429

476 NCON K2c59

477 NCON K68

478 NCON K8233

479 NCON K29c

480 NCON Kefb7

481 NCON K3f0

482 NCON Kfe6f

483 NCON K1eb

484 NCON Kee26

485 NCON K5dc

486 NCON Kfcde

487 NCON K3d7

488 NCON Kb95

489 NCON K1d3

490 NCON Kfb4d

491 NCON K5c3

492 NCON Ka05

493 NCON K3bf

494 NCON K18bc

End of 36 months entry

495 NCON K1ba

496 (NOP)